Manual

Machine Data Management

MDE-MDM 8.2

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Contents

2 Overview: Machine Data Management

Summary

Purpose

Machine Data Management provides functions that make it possible to automatically transfer unit quantities and machine statuses from connected machines / systems. Depending on the type of connection, there is the ability to monitor the planned cycle time accounting for predefined machine tolerances.

If automatic transfer to certain machines is not possible, you can establish a semi-automatic connection or enter data manually.

A full range of configuration and validation checking functions make it possible to adapt precisely to each machine and to control the data collection process.

Implementation notes

You use Machine Data Management if

- You would like to transfer unit quantities and statuses provided from machines and systems into the system automatically.
- You would like the option to either manually enter machine statuses as they occur or have it automatically entered with the support of the system.

Integration

Machine Data Management is integrated into numerous other system function packages:

- Detailed scheduling / shop floor scheduling
    - Recorded machine statuses are visualized online in shop floor scheduling on the planning board so that the scheduler is given an overview of the current production situation.
    - Depending on system configuration, remaining run time is calculated based on the cycles or quantities recorded.
- Graphic machinery
    - In graphic machinery the recorded machine statuses and cycles / unit quantities are visualized online.
- Tool / resource management

  - In tool / resource management, maintenance activities can be triggered and planned based on the numbers of units recorded.

Features

- Machine management
  - Database configuration tables for workplaces and machines
  - Individual configuration of up to 9,999 states/ statuses per machine or workplace; grouping of statuses into status hierarchies
  - Print function for creating a status list with barcode
  - Definition of status classes that are posted to the machine states / statuses upon occurrence
- Status recording
  - Manual recording of machine states/ statuses at BDE terminals
  - Extensive validation checks (e.g. check whether setup status may be set if no order is logged on)
  - Automatic recording of machine states/ statuses at BDE terminals
- Entry of quantities
  - Configuration defining logical counter inputs for the automatic recording of production figures
  - Automatic recording of production figures (cycles, yield and scrap) at BDE terminals
  - Manual recording of production figures (cycles, yield and scrap) at BDE terminals
- Machine monitoring
  - Monitoring of the planned cycle time accounting for predefined machine tolerances
  - Machine monitoring based on operating signals
  - Suppression of the production status despite active production signals (e.g. during setup status)
  - Explanation for disturbances/malfunctions based on machine monitoring
- Posting:
  - Posting of machine status times to machine-related log records (postings)
  - Posting of machine status times to status classes
- Posting maintenance
  - Display of machine-related log records (postings) generated based on entered data
  - Option to edit and correct machine-related log records

3 Data Collection in HYDRA MDE

3.1 Summary

The posting principle of HYDRA MDE corresponds to that of HYDRA BDE: once again the collection and processing of dialogs constitute the basis. Processing of a dialog results in events that constitute the basis for the updating (calculation of statuses) and posting (generation of log records).

Dialogs

As input values of data collection - dialogs represent the system's interface with the shop floor.

Events are generated by resolving the dialog. The dialog is resolved into individual events based on dialog data, the processing logic of the dialog (incl. configurations) and the application of the dialog according to the context (posting status).

HYDRA MDE input dialogs are among other things:

Dialog	Meaning
M_MST	Change (machine) status
M_SZY	Change target cycle (with monitoring based on cycles)
M_TLG	Change partitioning/cavity
M_AST	Cyclic status update is automatically generated by the terminal; cannot be recorded manually

Events

Events result from dissolving dialogs and are thus the result of the input. Which events get to the system at what times is an important point of controlling the software.

Further, partly optional, input functions are:

→ logging of the dialog and the resulting events,

→ provision of the posting result,

→ provision of current information in the form of lists (e.g. machine lists),

→ escalations (with M_MST).

4 HYDRA MDE Log Records

General

HYDRA-MDE log records describe postings based on machines/workplaces. A HYDRA-MDE log record is an evaluated data record that is generated due to posting events. An MDE log record documents among other things:

- Period of time (beginning, end) for which the log record applies,
- Duration since the last status change,
- Created status,
- Resource performance account which the status is assigned to,
- Target cycle that was set when the status changed (end of log record),
- Partitioning that was set when the status changed (end of log record),
- The number of cycles that have been recorded within the period of the log record,
- Computed quantities for meter readings that have been recorded within the period of the log record.

HYDRA-MDE log records do not have a direct relation to HYDRA-BDE log records. Thus, the period of an MDE log record does not depend on HYDRA-BDE log records. In exceptional cases, there might even be MDE log records without that an operation was logged on during that period of time (and as a result, without generating HYDRA-BDE log records for it). Which MDE log record is generated at what point in time depends, in particular, on the triggering posting event. The different log record types are distinguished by their record type. The machine-related record types that are described in the sections that follow are to be distinguished.

Record type P

General

A log record of the record type P is generated when the workplace/machine status is changed. It has been configured to be able to evaluate the period of time when a status occurred.

Triggering events and dialogs

Events: Machine status change (M_MST)

Dialog: Machine status change (M_MST),

Special remarks

The quantities included in an MDE log record are the quantities posted in this period of time. These quantities result from manual quantity postings (partial uploads A_TR) or from automatically recorded meter readings which might have been converted using conversion factors (partitioning, pulse factor).

The values in the "counter" fields do not include deltas, but quantities and cycles that have been added up since the beginning of the shift. The delta quantities (as of HYDRA-MDE 7.2) are provided in separate fields for evaluation purposes.

Record type N

General

A log record of the record type N is generated when the shift ends (automatic shift change, requires a HYDRA-MDE machine). It has been configured to evaluate the period of time prior to the shift end when the status already existed.

Triggering events and dialogs

Events: Machine status change (M_MST)

Dialog: End of shift (A_AAB)

Special remarks

The notes given for record type "P" also apply in this context.

5 Posting of Times

The machine duration (or duration) is determined by the time interval between two status changes. It is synchronized with the shift calendar of the machine, and planned shift breaks are removed from the time interval calculation.

Please note:
The determination of durations when status messages/postings are recorded online at the terminal (or via PDM) only takes into account a limited number of shifts. This relates to the last point in time postings were made for this machine.
However, durations might be missing or incorrect in MDE log records if no postings (status changes, shift changes, or similar) are entered for a machine over a longer period of time (e.g. the terminal is shut down).

Special features

The following configuration option may influence the above-mentioned posting of times/durations:

Post production time to main utilization time (MUT) during break
Cross-system configuration in the basic parameter settings of HYDRA.

Waiting period processing, machine
Cross-system configuration in the basic parameter settings of HYDRA.

Automatic status change to status 999
Shift-related configuration at the HYDRA-BDE day type.

6 Booking of Quantities

Data collection in different quantity types and quantity accounts

At the machine or workplace, you can collect the quantities in different quantity types and for different quantity accounts.

The following **quantity accounts** are supported:

- Yield
- Scrap
- Rework
- Open quantity (problem quantity)

The following **quantity types** are supported with each quantity account:

- Primary quantity
- Secondary quantity
- Tertiary quantity
- Base quantity

The automatic collection of quantities on the terminal always refers to the primary quantity.

Conversion to alternative quantity units

Bookings for alternative quantity accounts can be performed as follows:

- direct (manual) input
- conversion
    - from other quantity types with manual input
    - from primary quantity with automatically collected quantities

Direct (manual) input

If a quantity is directly (manually) entered in a quantity unit of a quantity type, an automatic conversion is not performed.

Conversion from other quantity types

If alternative quantity accounts are not collected, the server converts the quantities to alternative accounts using the conversion factors or quantity units, which are configured in the master data of the machine/workplace on the console.

In general, conversion first takes place into the base quantity unit (unless this one is recorded manually) and from the base quantity unit into the alternative unit (unless this one is recorded manually).

Identical quantity units

If quantity units are identical they are converted via numerator and denominator in the master data of machines/workplaces. If numerator and/or denominator = 0, then the quantities are taken over 1 to 1 without being converted.

Different quantity units

If quantity units are different, conversion is performed in the following order of priorities:

- Conversion using the numerator and denominator defined in the workplace/machine master data (always convert into base quantity unit first, then into the alternative unit);
  If numerator and/or denominator = 0

- Conversion using the formulas of the quantity units .

No quantity units

If quantity units are not assigned for the machine, no conversion is performed.

Conversion of quantity 0

A quantity 0 is generally not converted into alternative units, even if a value that is not 0 could be calculated (e.g. using a formula).

Quantity conversion of automatically collected quantities

To convert automatically recorded quantities using formulas, you can use
- fixed factors/values or values based on machines/workplaces (user fields);
- data that is specific to the operation, such as length, width, weight per piece, etc.

You use the operation logged on the longest to identify the OP-specific data. Consequently, the following (logical) restriction arises for operations that are logged on at the same time when it comes to quantity conversion:
- the operations must produce the same material;
- the operations must have the same default data (length, width, weight per piece).

Any further requirements must be taken into account as part of customer projects.

Basis for HYDRA-MDE quantity conversion

In the *Workplace/machine configuration*, tab *Configurations > Quantities*, you can use the configuration option *Basis for HYDRA-MDE quantity conversion* to use the configured quantity conversion of the running operations also for the machine. This option ensures a correct calculation of quantities even if more than one operation is active :
M – conversion factors of the workplace (APZ) [default]
A – conversion factors of the OP if logged on, otherwise workplace

Notes

If configuration A and different quantity units of operations are used, the quantities are accumulated and booked to the machine accounts without reference to the units.

When you edit postings, the MDE-related quantity conversion using the operation data or the machine/workplace values (user fields) is not supported. In this case, do not enable the option *Convert quantities* in the editing dialog of the maintenance of postings.

Display of alternative quantity units on the terminal

The (manual) collection and display of alternative quantity units is only possible on Windows terminals (requires customization). Note: The terminal itself does not perform any local conversion into alternative quantity units; the quantities are only displayed when data is reloaded from the HYDRA server.

Automatic collection of quantities

You configure the counters for each machine. The following options are available:

- Posting of yield, scrap, rework, open quantity, no posting
- Posting of cycles (strokes)
- The quantity is calculated using partitioning (parts produced per cycle) and/or pulse factor
- Cycle monitoring
- Reason (e.g. scrap reason)
- Offset against “quantity account"
  e.g. scrap is deducted from yield

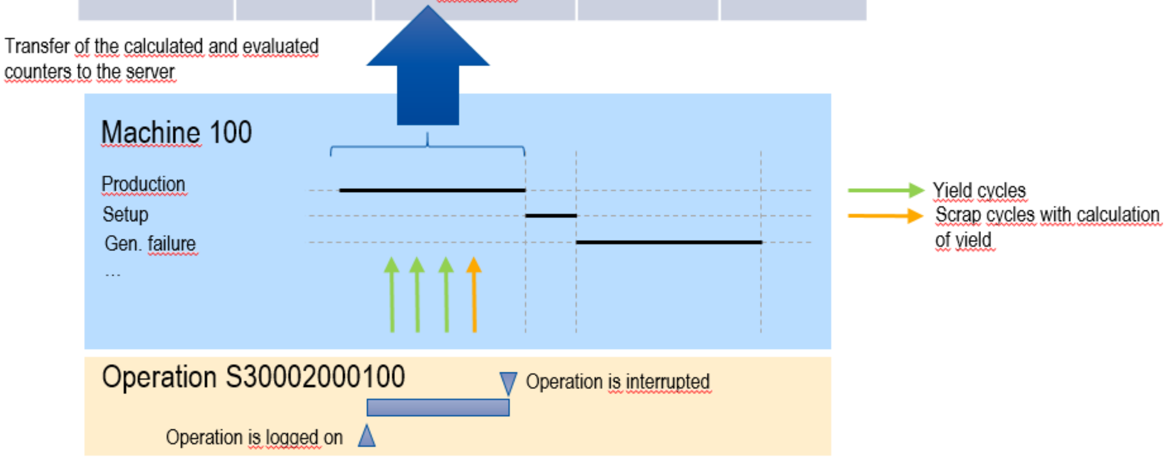
Quantities that are issued by automatically recorded counters are always posted in primary quantity. The conversion into other quantity types is described in the previous section. The HYDRA server can use different quantity accounts to calculate a quantity (e.g. to record the total quantity).

If the cycles are collected, not every single cycle is transferred to the server, but the collected and also the calculated cycles and the evaluated counters are cyclically transferred to the server. The values transferred are then integrated into the events or into the calculated quantities of the machine-related postings (MDE log records).

Example 1:
A yield counter and a scrap counter are defined for the machine. The scrap collected is offset against the yield collected.
The events and postings include the offset and calculated quantities. Not for each cycle, a posting is created.

Postings for orders

Order/OP	Workplace	Record type	Yield (P)	Scrap (P)
S30002000100	M100	Part quantity posting	2	0
S30002000100	M100	Part quantity posting	0	1
S30002000100	M100	Order interruption	2	1

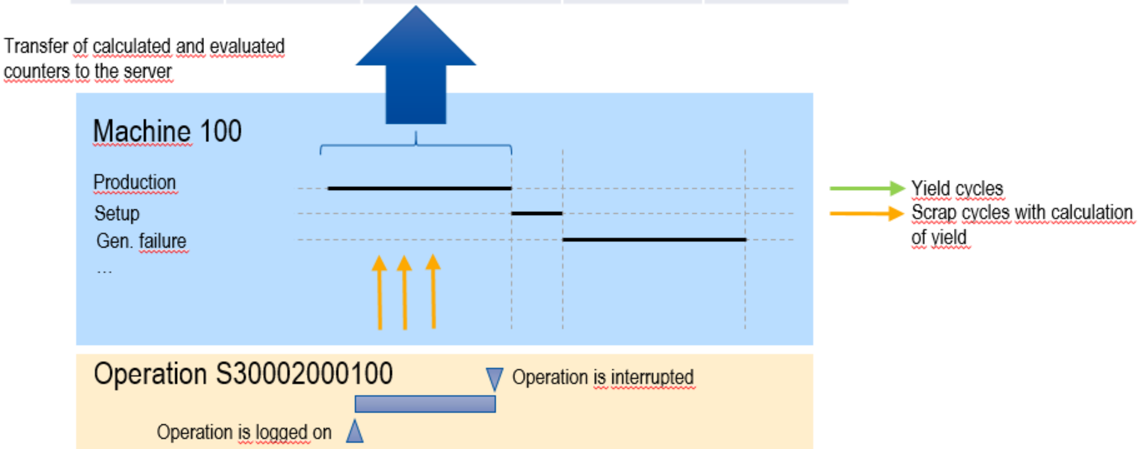


Example 2:

A defined calculation of a counter reading can also result in negative quantities in the posting records:

Postings for orders

Order/OP	Workplace	Record type	Yield (P)	Scrap (P)
S30002000100	M100	Part quantity posting	-3	3
S30002000100	M100	Order interruption	-3	3



Counters that are configured with the "no posting" option do not post any quantities to a quantity account. Using these counters, the following use cases can be integrated:

- Cycle monitoring only

- Cycle monitoring and posting as cycles (strokes)
- Posting as cycles (strokes) without cycle monitoring

Quantity calculation and parallel make-to-order production

With operations that are logged on at the same time, quantities are posted with respect to the order according to the relevant specifications (partitioning, pulse factor) of the operation.

This specific calculation of quantities is performed with all quantity accounts that are recorded automatically (yield, scrap, rework, open quantity). Counter pulses or quantities resulting from this calculation are generally posted onto *all* OPs that are logged on (according to the configuration: with activated partitioning/pulse factor or not).

Specifications are identified as follows:

Partitioning/cavity (TLG)

Partitioning of the machine (parts per cycle): $(TLG_{OP1} + TLG_{OP2} + TLG_{OPn}) * TLG_{Machine}$

If a partitioning is specified for a machine, the partitioning of the operations and the machine partitioning are multiplied.

If you interrupt/log off an operation, the partitioning of the machine is updated, i.e. the total of the remaining operations is recalculated.

If the last operation is interrupted or logged off, the partitioning is reset to 1 and multiplied by the machine-specific partitioning.

Partitioning of the operation: $TLG_{OP} * TLG_{Machine}$

Changing the partitioning on the terminal

If the partitioning of the operation is changed, the partitioning of the machine and of the operation is changed/updated.

Pulse factor (IMPFAKT)

Pulse factor of the machine = minimum($IMPFAKT_{OP1}$, $IMPFAKT_{OP2}$,… $IMPFAKT_{OPn}$) * $IMPFAKT_{Machine}$

Pulse factor of the OP = minimum($IMPFAKT_{OP1}$, $IMPFAKT_{OP2}$,… $IMPFAKT_{OPn}$) * $IMPFAKT_{Machine}$

Note:
The same pulse factor applies for all active operations. Therefore, you must ensure that parallel operations get the same default pulse factor.
This means: The same pulse factor is used for all operations when quantities are calculated.

Quantity calculation on the basis of partitioning and pulse factor

Quantity of the machine = <number of cycles> * partitioning of the machine / pulse factor of the machine

Quantity of the operation = <number of cycles> * partitioning of the operation / pulse factor of the operation

Note:
The pulse factor is calculated as a fraction. When the quantity is calculated, the pulse is used as denominator and the partitioning is the numerator.

Display on terminal

On the terminal, the field **Partitioning** shows the factor that is relevant for the quantity calculation of the machine. It is

partitioning of the machine / pulse factor of the machine.

As described above, the different default values of the machine (e.g. machine-specific partitioning) and of the active operations are used to calculate this factor.

On the terminal, the field **Target cycle** shows the relevant current target cycle (the cycle extension is not integrated). This is max($SZY_{OP1..n}$) for OPs that are logged on at the same time.

Output "target quantity reached"

The target quantity output of the machine interface, e.g. setting a lamp, is set when the smallest target quantity of a logged on order is reached. If the OP with the smallest target quantity is interrupted or finished, the next OP with the smallest target quantity is used.

Manual collection of quantities

You can also use the configurations below to book manually recorded quantities:

- Offset quantities (accounts) against other quantity accounts (*Allocation with* option) e.g. deducting manually recorded scrap from yield

- Post manual quantities as cycles

Offsetting against another quantity account

You can offset automatically and manually recorded quantities against another quantity account, e.g. you can deduct the manually recorded scrap from the yield.

You can use the *Allocation with* option for

- automatically recorded quantities in the counter configuration

- manually recorded quantities in the machine configuration.

If you use these options, bookings (BDE log records, MDE log records) with negative quantities or negative order quantities can result.

7 Workplace and Resource Configuration

Overview

HYDRA menu	Master data → Resources → Resource configuration Master data → Workplaces/machines → Workplace configuration
FEDRA menu	Detailed Scheduling → Master data → Resource configuration
Transaction code	res
Function authorization	mdres mdresgenh for fields in combination with Test Equipment Management

Available user fields		
Where?	**Object type/user field key**	**Source (type)**
Tab *User fields*	<Res.type*)>/depending on data record	Resource (MF-D)
Table	RES/SYSTEM	Resource (MF-D)

*) <Res.typ> = resource type

The resource configuration is the central function to manage resources in the MES.

Purpose

This application manages the master data of workplaces/machines and other resources (tools, DNC resources, etc.). The resource type classifies resources. Each resource type is also linked to specific functions and applications, which provide further functionalities of the MES for resources of the specified type.

Integration

Use this application to view the resource information of all resource types available in the system. The resource type also specifies how and if data records can be edited. Depending on the resource type, you cannot edit all fields or create and delete all resources.

Based on the resource type, the MES also includes further applications that are especially tailored to these types. The *machine data collection* application package, for example, is based on resources of the type "machine".

In addition to the resource configuration, the *resource overview* application is available. You cannot use the *resource overview* application to edit data. This application only allows administrative operations for the daily handling of resources such as the stock transfer of resources.

Requirements

Create a year model/shift calendar prior to creating a workplace or machine. If you want to use the various resource types effectively, you also need the advanced licenses for these types.

Selection criteria

The application provides the following selection criteria:

Resource from ... to ...

This selection criterion refers to the resource. You can also use wildcards (placeholders *).

Short name

Short name of the resource. Only relevant for resources of type MNR.

Resource type

Type of resource.

Workplaces and machines always have the resource type MNR. But you can assign individual resource types to the other resources by configuration. Predefined resource types include:

DNC	NC/DNC program
DOC	Document
ENE	Energy meter
ENT	Removal device
ENT	Removal device
MNR	Workplace/Machine
PAC	Packaging, transportation container
PRM	Test and measuring equipment
PER	Production staff / general
PRU	Setup staff
TEM	Tempering equipment
VOR	Device
WNR	Tool

We recommend using the predefined resource types.

The displayed detail resource information varies with the resource selected in the table overview.

Name

Name of the resource.

Group

Workplace/machine group of the resource. Only relevant for resources of type MNR.

Cost center

Cost center of the resource.

Short name

Short name of the resource.

Resource family

Family the resource is assigned to.

Responsibility area

Responsibility area the resource is assigned to.

Storage location

Regular storage location of the resource.

MD user fields

MD user fields 1- 6 of the resource. If you select a resource family in the selection panel, the application shows the field names according to the assigned user field definition.

Field descriptions

This detail application includes four main tabs:

- Resource configuration
- Resource list
- Resource attributes
- DNC versions

Main tab *Resource configuration*

Here, you can define the configurations and master data of resources.

General tab

Resource type

Resource type of the resource. The system delivery includes some default resource types. Create additional resource types in the application .

Resource

Enter the number of the resource or workplace to be collected in this field.

The resource type also specifies the maximum number of characters that are allowed for the resource number:

- Resources of the type MNR: a maximum of 8 digits
- Resources of a type <> MNR: a maximum of 20 digits

Permitted characters: ABCDEFGHIJKLMNOPQRSTUVWXYZ1234567890_.-+#. Do not use spaces and other special characters. For technical reasons, you can enter * (asterisk) and % (percent), but they are nonetheless not permitted because they are not valid characters. When you exit the input field, the system automatically converts lower case letters into CAPITAL LETTERS.

Please note for workplaces/machines (resource type MNR):
For technical reasons, the system does not check the maximum number of digits allowed for resources of the type MNR. For this reason, make sure that the resource number length (= workplace/machine number) does not exceed 8 digits.
Please note: If you set the resource type MNR before entering the resource ID (machine number), the GUI only allows you to enter eight digits.

If you select the option "numeric machine number" (basic parameter settings) for use with DOS terminals, you must ensure that the resource number (= workplace/machine number) only includes numerical digits and that its length is exactly 8 digits. If necessary, prefix leading zeroes to the number to extend it to eight digits, when creating the workplace/machine.

Short name

Short name of the resource. Only use this field with workplaces/machines (resources of the type MNR).

Name

Use this field to assign a short, unique name to each resource. Reports and overviews as well as terminal dialogs show this name, which is also useful for orientation purposes.

Responsibility area

Use responsibility areas to restrict the data users can view in different evaluations/reports. Users can only view the data they are allowed to according to their responsibility area authorization.

The *responsibility area* field can also remain empty. In this case, the resource is always displayed regardless of the user's assigned responsibility authorizations.

If you leave the *responsibility area* field empty, the system automatically enters the value "--DEFAULT--" in the field. Resources including this value are always displayed regardless of the user's assigned responsibility authorizations.

Cost center

This field includes the cost center the resource is assigned to.

Inventory number, engraving number, drawing number, manufacturer, owner

Additional information in form of comments.

Acquisition date, acquisition costs

Additional information in form of comments.

Configure the currency for the entire system in the basic settings.

Storage location

Location where the resource is stored when it is not being used (original storage location).

In connection with the Material and Production Logistics (MPL) product group, this field specifies a material buffer. If you log on an input batch, the logged on input batch(es) will be transferred from the previous material buffer to the material buffer entered in this field (upstream of the machine).

Delivery date, start-up date, guarantee date

Additional information in form of comments. These fields are only available if Test Equipment Management (PMV-PPK or PMV-SVP) is licensed and the right "mdresgenh" is assigned.

External designation, resource type designation, usage, purchase order number

Additional information in form of comments. These fields are only available if Test Equipment Management (PMV-PPK or PMV-SVP) is licensed and the right "mdresgenh" is assigned.

Supplier and party in charge including detail fields

Additional information in form of comments. These fields are only available if Test Equipment Management (PMV-PPK or PMV-SVP) is licensed and the right "mdresgenh" is assigned.

Workplace configuration tab

This tab is only available if you select a resource of the type "MNR".

Workplace master data

Workplace category

N Machine

P Workplace

Defined as machine or workplace. If you exclusively use BDE and/or MDE and PDV, the two categories are identical as regards processing.

J Machining center (BDE-BEA only)

The "Machining center" category and its functionality are described in detail in the BDE-BEA product documentation.

L Line (MDE-SFL only)

A Aggregate (MDE-SFL only)

The categories "Aggregate" and "Line" and their functions are described in detail in the MDE-SFL product documentation.

Q CAQ inspection station

Workplace is defined as mere CAQ inspection station and does not affect BDE or MDE statistics.

R Coil-based manufacturing (only for coil-based manufacturing)

This type controls specific functions for the coil-based manufacturing.

S Cutting unit (only for coil-based manufacturing)

This type controls specific functions for the coil-based manufacturing.

D Parallel output batches (only MPL)

You can produce parallel output batches on the machine for an operation that requires batch management.

C Packing station (only MPL)

You can use specific posting functions of the machine to represent a packing station. The functions are described in detail in the AIP-LCS product documentation.

M Melting aggregate

This option defines a machine as melting aggregate in terms of composition.

F Laboratory/in-production inspection

This workplace is configured as inspection station. The inspection points are displayed, which are assigned to this workplace or machine group of this workplace because of the higher-level inspection point. You must activate the workplace-specific layout here. Use the following parameters for activation in the AIP layout file "globaldefines.xml".

<MachineSpecifiedLayout>True</MachineSpecifiedLayout>

W Goods receipt inspection

This workplace is configured as inspection station. The goods receipt inspection points are displayed, which are assigned to this workplace or machine group of this workplace. You must activate the workplace-specific layout here. Use the following parameters for activation in the AIP layout file "globaldefines.xml".

<MachineSpecifiedLayout>True</MachineSpecifiedLayout>

K Calibration

This workplace is configured as inspection station. The calibration inspection points are displayed, which are assigned to this workplace or machine group of this workplace. You must activate the workplace-specific layout here. Use the following parameters for activation in the AIP layout file "globaldefines.xml".

<MachineSpecifiedLayout>True</MachineSpecifiedLayout>

Workplace type

E Single workplace (SWP)

G Group workplace (GWP)

Group workplaces are workplaces without machine data collection or MDE evaluations.

In case of group workplaces, you cannot post to resource performance accounts in an operation-related manner with postings based on the current machine status. Only main production times (RPA 11) are recorded. You must define a status with the control indicator "production" in the .

The system does not generate for group workplaces. Therefore, MDE evaluations that evaluate MDE log records are not possible.

Like single workplaces, you can assign group workplaces to terminals. In this case, you have to make sure that the is set to operation mode "BDE" or the option *Processing* is set to "BDE processing" in the .

External workplace

This field identifies external workplaces. Currently, it only functions as a comment.

Locked

If this option is checked, the machine/workplace has been (logically) deleted. In this case, the system does no longer permit the following changes:

- Order postings on the terminal
- Order postings on the MOC (e.g. using the "order overview" function)
- Changes when editing events

The graphic planning board of the Shop Floor Scheduling and the application *Workplace assignment* do no longer show the machine/workplace.

Blocked machines/workplaces are shown in evaluations and overviews. If blocked machines/workplaces are not shown, this is then described in the relevant documentation of the MOC application.

Tip: In applications where data is selected according to the responsibility area authorization, you can hide machines/workplaces if you remove the responsibility area.

Company

Use this field to differentiate the individual machines/ workplaces. The system can use this field for evaluation purposes.

Group

Use this field to assign the workplace/machine to a logical group. In planning, this is a capacity group. Capacity groups combine primary capacities.

If you create a new workplace, it is automatically assigned to a group of the same name (menu BDE: Master data > Workplaces/machines > Groups), which is defined as a capacity group. If the capacity group does not yet exist, the system automatically creates a capacity group and assigns the workplace.

Category

Enter the category of the machine. By means of this, you can enable a validation check according to the BDE configuration: *Master data > Order configuration > Order types,* tab *validation,* option *Check planned workplace/group/category on OP logon* (value *category).*

Year model

Enter a valid year model . The times to be posted are compared with this shift model when they are recorded. If you have not defined a planned year model in the HLS tab, the shift model entered here is also used in the Shop Floor Scheduling.

Standard rate, machine

Enter the arithmetical standard rate of machines for calculations. The Shop Floor Scheduling uses this value for some (evaluated) KPIs.

Standard labor rate

Enter the arithmetical standard labor rate for calculations. The Shop Floor Scheduling uses this value for the KPI "Evaluated labor utilization".

Performance level

You can enter the performance level of the workplace/machine in percent in this field. The Shop Floor Scheduling and the evaluation of material requirements integrate this value when calculating the remaining run time.

Incentive wage indicator

Defines the type of calculation used for incentive wages. This option is mostly used in combination with the incentive wages based on formulas for customer-specific configurations. In addition, use the "incentive wage indicator" as selection criterion for the *wage type determination* to calculate incentive wages.

Leave this field empty, if you do not use the incentive wage module.

The *incentive wages indicator G=group calculation* has a special meaning. If this option is set for a workplace/machine, you have to assign a premium group every time you log on an order. You can do this either via
- the "assignment of premium groups" option of the product group *Incentive wages* or, optionally, via
- an additional field in the terminal dialog for the logon of orders. If no assignment is available, the system rejects the logon of the order by issuing a validation error.

Therefore, you may only assign the incentive wage indicator *G = Group calculation,* if the group premium conditions are met in the incentive wages calculation, as otherwise orders can no longer be logged on!

You can specify the meaning of the other incentive wage indicators according to your requirements while customizing the system.

File

You can assign a graphic to each machine/workplace. The *workplace overview* or the AIP shows this graphic, for example. The following image formats are supported: jpg, gif, tif, bmp, ico, emf, wmf.

In the path configuration, you must have configured the following:
- the path ID "MOCWPIMG" for the MOC or SMA
- the path ID “HYDRA” (also see) for the AIP. The file name length of graphic files is restricted to 12 characters (8.3 notation). Note for Linux installations: only use lower case letters for file names.

Maximum capacity (KG)

If a machine is configured as melting aggregate, define the maximum capacity in KG here.

Accuracy class, unit, etc.

Information fields in order to describe the accuracy. These fields are only available if Test Equipment Management (PMV-PPK or PMV-SVP) is licensed and the right "mdresgenh" is assigned.

Data collection

Display 3rd list

Use the options described here to show/enable a third list in the main view of a Windows terminal (CTWIN / AIP). You can switch between the respective terminal lists depending on the options set. The following settings are possible. Please note that the contents displayed in the lists depend on the product group in use:

→ Input material (MPL): shows logged on input materials/ batches.

→ Resources (WRM): shows logged on resources and tools.

→ Staff (BDE): shows logged on staff.

→Output material (MPL): Produced output batches are displayed.

Show material/PRT list when OP is logged on

This option is only relevant in connection with the WRM module and the resources logged on to the Windows terminals (CTWIN / AIP).

If this option is set and you log on an OP, a specific login dialog opens. This dialog includes a list of components/production resources and tools. This list shows resources that meet at least one of the following requirements:
- the option "posting to terminal" is set in the resource type;
- the option "log on with OP" is set to "explicit logon" for the resource.
- the resource is a so-called "required resource" (option is set for the resource).

Please note: If the workplace is relevant for MPL, the list also shows material components.

Sequencing list

This option defines which operations are displayed in the sequencing list of the terminal. The following settings are available:

S Basic setting. The system takes the value from the option of the same name in the *HYDRA basic settings*.

M Pool of workplaces. The terminal sequencing list only shows the operations planned for the workplace.

G Pool of workplaces and groups. The terminal sequencing list shows operations that are:
- planned for the current workplace or
- for another workplace of the group or
- that are still located in the pool of groups.

K Pool of workplaces and categories. The terminal sequencing list only shows the operations that are planned for workplaces of the selected category.

H Group control. The terminal sequencing list shows the operations that are
- planned for the current workplace or
- for another workplace of the group.

Number of OPs in sequencing list

Enter the maximum number of operations that are to be displayed in the terminal sequencing list. Enter 0 if you want to show all operations.

Compulsory sequence

Use this option to specify if it is mandatory to log on the OPs in the planned sequence. The following parameters are permitted:

N Disabled
J Enabled

If the parameter is "enabled" and you log on an OP, the system checks whether the order backlog for this machine/workplace includes an OP that is planned for the same time or previous to this OP, but has not yet been started (i.e. status = V/prepared). If yes, the system rejects the logon of this OP.

Note: If you plan orders in the system using the *Order sequencing* (menu Production control → Production support → Order sequencing) and you configure the sequencing list with any other option than "M" (pool of workplaces) and you enable the *compulsory sequence,* this might lead to a combination that does not make sense.

Please note for the sequencing list:

- If the sequencing list includes operations that are in the status "interrupted", you can log on these OPs at any time, irrespective of the specified compulsory sequence.

Dialog control

To meet this requirement, define a dialog control that deviates from the standard behavior for the workplace in the dynamic dialog configuration of the Windows terminal (CTWIN / AIP). Then refer to the dialog control in the dialog.

Use this configuration only as part of customizing the HYDRA system. Otherwise the configuration is not relevant.

Logon of several OPs

Select this option, if several different operations should be processed on the machine. Otherwise, the system only allows one operation to be logged on to the machine.

Possible values:

Y	Log on as many OPs as required at the same time. Please note: The system allows a maximum of 20 operations to be logged on simultaneously to a machine, if the machine is assigned to a terminal with operation mode *MDE.* If more than 20 operations must be logged on at the same time, MPDV must review the conditions in order to remove the limitation. If MPDV agrees to remove the limitation, you can do so, otherwise search for alternative solutions. MPDV analyzes the conditions as part of a service.
N	You can log on one OP only.
1...9	You can log on a maximum of n OPs.

Posting

Quantity posting to staff

Use this function to post the quantity of order interruptions/logoffs to the person who is logged on for the longest period.

Detailed information about quantity posting to staff can be found .

Posting for OPs that are not logged on

Use this option if you want to
- interrupt
- finish
- report part quantities for
operations that are not logged on to this workplace.

If you record quantities for an operation that is not logged on, the system posts these quantities onto the operation in the BDE module. The MDE module does not post the quantities.

If you want to use this function with the AIP terminal, the BDE posting dialogs that are installed by default require the following:
- use the simplified BDE posting dialogs (the so-called "") or
- customize the dialogs.
Then you will be able to enter an operation that is not logged on.

Posting of machine time with simultaneously logged on operations

If this option is set and OPs are logged on simultaneously, the system posts the machine time proportionately onto the operations.

Y	Proportionate posting on OP according to the number of OPs
N	No proportionate posting. If the option is not set, the complete machine time is posted for each operation.
V	According to the default quantity of the OPs. Make sure that the default quantity (target quantity in primary quantity unit) of the operation is > 0.
Z	According to the standard time of the OPs. Make sure that the standard time (processing time) of the operation is > 0.

Please note:

This option is also evaluated for group workplaces and in general you should better not use this option for group workplaces.

Automatic logoff of staff when shift ends

This option is only relevant, if you set an "X" for (enable) the option of the same name in the order type.

Use this option to configure the personnel-related data collection at MDE workplaces. If you use HYDRA MDE, the terminals can generate fully automatic shift ends. You can configure here if
- the staff logged on to the workplace should be logged off automatically at the end of the shift or
- if they should remain logged on.

Y	Always log off staff when the shift ends.
N	Always save staff when the shift ends except for manual logoff.
X	Evaluate the person's settings. The system searches for the corresponding settings of the person .

Automatic OP posting when shift ends

This option is only relevant, if you set an "X" for (enable) the option of the same name in the order type.

Y Interrupt and log on again at beginning of shift

N Interrupt

Shop Floor Scheduling

Find further information about the HLS product group in the relevant HLS documentation.

Planning function

This option specifies whether a workplace or a machine will be displayed and if so, in which planning function.

P Planning in the graphic planning board of the Shop Floor Scheduling or in the graphic order sequencing (GAV), i.e. you plan the workplace via the Shop Floor Scheduling or the graphic order sequencing; the workplace is then displayed in these applications, but not in the tabular order sequencing (AVG).

Note: There are also other settings that specify whether a workplace is displayed in the Shop Floor Scheduling or in the graphic order sequencing:
- the workplace must be assigned to a group identified as a "capacity group"
- you must be authorized for the responsibility area of this workplace
- planning profile

H Only relevant, if you use the HYDRA Shop Floor Scheduling module (HLS).

Like P.

T Reserved

A Planning in the tabular order sequencing (AVG), i.e. you plan the workplace using the AVG product group.

N No planning; the tabular order sequencing (AVG), the graphic order sequencing and the HLS module do not show the workplace.

Planned year model

Here, you can enter a special year model only used for planning in the Shop Floor Scheduling. This year model does not affect data collection and posting in the product groups BDE/MDE. If you do not define a planned year model, the system uses the year model (*Master data* tab) for the planning.

Availability

Define the available capacity of a workplace/machine. The default value for the available capacity is 1000 [per mill].

In the Shop Floor Scheduling, the capacity check and automatic assignment assume that each operation has a capacity requirement of 1000 [per mill], i.e. exactly one operation can run on the workplace/machine at a time. In case of a manual multiple assignment, a dialog informs you about the double assignment. If you use the automatic assignment, multiple assignments are generally not feasible.

Use this setting to extend the availability of the workplace such that a multiple assignment is permitted. If the workplace capacity allows, for example, processing of two operations at the same time, set the available capacity to 2000 [per mill] in this field.

If nothing is entered in this field or if you enter the value 0, the system interprets this as the default value of 1000 [per mill].

This functions requires a corresponding license.

Check personnel availability

Choose from the following options:

- Check if at least one person is planned
- Check personnel availability
- Check personnel availability and qualification

When you operations in the , the system checks whether persons are planned in the application for the time of the scheduling You will find further information on the display of personnel capacities in the Graphic Planning .

This option is only available if you enable the extension .

MPL

For further information on the MPL product group, refer to the relevant MPL documentation.

Batch management

Activates the entry of the batch number for this machine within the terminal posting dialogs. Possible values are:

N No batch processing

L Batch tracing (input/ output batches) as part of HYDRA MPL/TRT

D Throughput batch processing as part of HYDRA MPL/TRT

J Individual batch tracing (CHV)

The following functions are only available in connection with the product group Material and production logistics and are supported only by Windows terminals (CTWIN / AIP).

Preceding material buffer

Irrelevant.

Subsequent material buffer

If you specify a material buffer in this field, the field *Target buffer* in each of the entry dialogs (e.g. output batch change, log off operation) is automatically populated with this value.

If you do not enter a material buffer in the input dialog (e.g. deleted from the input field), the system automatically posts the output batch to the material buffer specified in the "subsequent material buffer" field.

Automatic generation of batch number

If you set this option, the system automatically generates a batch number for the output batch to be produced. Otherwise, the system expects you to enter the batch number for the new output batch to be produced, when you log on an operation or change the output batch.

Please note: If, in the field *Batch management* you set the option D (= *Throughput batch recording*), the system automatically sets the value for the *Automatic generation of batch number* to "J". In this case, you cannot enter the batch number manually.

Consumption balance

When you log off an OP, the system opens an additional dialog (V_BLZ) displaying the material components and their consumption quantities in relation to the OP that is currently logged on. In this dialog, you can also log off input batches that are still running. This option is only activated, once you have enabled the consumption balance for the material type of the output material.

Generate transport order for output batches

This option creates a transport order relating to batches for a generated output batch. The transport starts from the material buffer where the output batch is included. The configurations of the material type override the corresponding options of the resource configuration.

Generate transport order for input material

This option creates an article-related transport order relating to a material component, when you plan an operation for a machine via the Shop Floor Scheduling module. Transport starts from the output material buffer of the preceding operation. The configurations of the material type override the corresponding options of the resource configuration.

Quantities tab

This tab is only available if you select a resource of the type "MNR".

Conversion factors for base quantity

At the machine or workplace, you can collect the quantities in different quantity types and for different quantity accounts. In general, the system supports the following **quantity accounts:**

Yield

Scrap

Rework (Windows terminal CTWIN/AIP only)

Open quantity (problem quantity; Windows terminal CTWIN/AIP only)

The following **quantity types** are supported with each quantity account:

Primary quantity

Secondary quantity (Windows terminal CTWIN/AIP only)

Tertiary quantity (Windows terminal CTWIN/AIP only)

Basic quantity (Windows terminal CTWIN/AIP only)

The system design specifies the use of several quantity types or accounts. For example: If you want to enter the rework quantity manually, a corresponding input field must be configured in the input dialog (customization).

Use the quantity type "primary quantity" if you want to collect quantities automatically.

Quantity units and conversion factors for base quantity

Define a quantity unit for each quantity type. Use the alternative quantity accounts to enter data/quantities manually. In this case, the system does not convert quantities automatically.

If you do not enter data manually in the alternative quantity accounts, the server converts the quantities into the alternative accounts using:
- the conversion factors or
- the units that are configured in the MOC machine master data.

For further information on the conversion of quantities and examples, refer to the document .

Basis for HYDRA-MDE quantity conversion

Define the basis for the quantity conversion.

A Use the conversion factors of the OP that is logged on. If no operation is logged on, the system uses the quantity conversion stated in the machine/workplace configuration.

M Use conversion factors from the workplace configuration for the quantity conversion.

Units and conversion factors for base quantity (P)

Quantity unit (P)

Indicate the quantity unit you want to use for data collection at this machine/ workplace. If you collect quantities automatically, these quantities are generally primary quantities.

If you want to convert quantities automatically into another quantity type, indicate the conversion factors for the base quantity here.

Units and conversion factors for base quantity (S)

Quantity unit (S)

Indicate the secondary quantity unit you want to use for posting the quantities to the workplace/machine. If you want to convert quantities automatically, indicate the conversion factors for the base quantity here.

Units and conversion factors for base quantity (T)

Quantity unit (T)

Indicate the tertiary quantity unit you want to use for posting quantities to the workplace/machine. If you want to convert quantities automatically, indicate the conversion factors for the base quantity here.

Units and conversion factors for base quantity

Quantity unit (B)

Indicate the base quantity unit you want to use for posting quantities to the workplace/machine.

Manual entry of quantities, yield

Manual entry of yield

Set this option, if you want

- to collect quantities manually;
- to set off the quantities against another account;
- to post the manual quantities as cycles.

For Windows terminals this option does not affect the quantity fields displayed in the input dialogs. Change these quantity fields via dialog configurations (terminal configuration or customizing of the dynamic dialogs).

Allocation of yield

Requirement: Set the option "Manual entry".

Use this option to offset manually entered quantities against other quantity accounts. In this case, the entered quantity is deducted from the specified account.

Note: If you offset quantities, the resulting values can be negative values.

Note

Do NOT set this option for DOS terminals, if in the counter configuration yield is offset against scrap or scrap is offset against yield.

Posting yield as cycles

Requirement: Set the option "Manual entry".

If this option is set, the system also posts manually entered quantities as cycles. Note here that the entered quantity is posted directly as cycles (partitioning is not integrated).

Manual entry of quantities, scrap

Manual entry of scrap

Set this option, if you want
- to collect quantities manually;
- to set off the quantities against another account;
- to post the manual quantities as cycles.

For Windows terminals this option does not affect the quantity fields displayed in the input dialogs. Change these quantity fields via dialog configurations (terminal configuration or customizing of the dynamic dialogs).

Allocation of scrap

Requirement: Set the option "Manual entry".

Use this option to offset manually entered quantities against other quantity accounts. In this case, the entered quantity is deducted from the specified account.

Note: If you offset quantities, the resulting values can be negative values.

Note

Do NOT set this option for DOS terminals, if in the counter configuration yield is offset against scrap or scrap is offset against yield.

Posting scrap as cycles

Requirement: Set the option "Manual entry".

If this option is set, the system also posts manually entered quantities as cycles. Note here that the entered quantity is posted directly as cycles (partitioning is not integrated).

Manual entry of quantities, rework

Manual entry of rework quantity

Set this option, if you want
- to collect quantities manually;
- to set off the quantities against another account;
- to post the manual quantities as cycles.

For Windows terminals this option does not affect the quantity fields displayed in the input dialogs. Change these quantity fields via dialog configurations (terminal configuration or customizing of the dynamic dialogs).

Allocation of rework

Requirement: Set the option "Manual entry".

Use this option to offset manually entered quantities against other quantity accounts. In this case, the entered quantity is deducted from the specified account.

Note: If you offset quantities, the resulting values can be negative values.

Note

Do NOT set this option for DOS terminals, if in the counter configuration yield is offset against scrap or scrap is offset against yield.

Posting the rework quantity as cycles

Requirement: Set the option "Manual entry".

If this option is set, the system also posts manually entered quantities as cycles. Note here that the entered quantity is posted directly as cycles (partitioning is not integrated).

Manual entry of quantities, open quantity

Manual entry of open quantity

Set this option, if you want
- to collect quantities manually;
- to set off the quantities against another account;
- to post the manual quantities as cycles.

For Windows terminals this option does not affect the quantity fields displayed in the input dialogs. Change these quantity fields via dialog configurations (terminal configuration or customizing of the dynamic dialogs).

Allocation of open quantity

Requirement: Set the option "Manual entry".

Use this option to offset manually entered quantities against other quantity accounts. In this case, the entered quantity is deducted from the specified account.

Note: If you offset quantities, the resulting values can be negative values.

Note

Do NOT set this option for DOS terminals, if in the counter configuration yield is offset against scrap or scrap is offset against yield.

Posting open quantity as cycles

Requirement: Set the option "Manual entry".

If this option is set, the system also posts manually entered quantities as cycles. Note here that the entered quantity is posted directly as cycles (partitioning is not integrated).

"MDE configuration" tab

This tab is only available if you select a resource of the type "MNR".

Monitoring

Monitoring type

Choose from the following monitoring types:

Monitoring via operating signal

No monitoring

Cyclic monitoring

If you select **cyclic** or **operating signal** monitoring, you can only enter a malfunction if the terminal prompts you to do so ("Assign malfunction"). If you do **not** use **automatic monitoring**, you can enter a new machine status at any time.

If you use the **cyclic** monitoring option, the machine automatically switches to the "production" status when counting pulses occur. If you select the "operating signal" option, the machine automatically switches to the status "production" as soon as the operating signal is set. If you do not use the "automatic monitoring" option, you must assign the "Production" status manually.

Entry of disturbance reason required with specified delay time in [s]

You can only use this function, if the following requirements are met:
- it is a Windows terminal (CTWIN, AIP)
- The Process Communication Controller (PCC) does not run in stand-alone mode.

If the system identifies a downtime without a reason, the terminal opens the input dialog "Change machine status" after the specified delay time. If the terminal goes back into production, the window still remains open.

If you now enter a machine status (during production), this data input activates a transfer posting event that changes the most recently recorded status from "General disturbance" to the newly entered status. If this change is ok, the window closes; otherwise, it remains open.

However, if the system identifies the next downtime (with or without a reason), you can no longer change to the previously noted status. The window closes automatically.

If the system identifies another downtime without a reason and the delay time has expired, then the input window opens as described above.

If the system identifies a downtime without a reason and the machine switches to production before the delay time expires, then the terminal does not automatically prompt you to enter a malfunction reason.

Important note:
This change only affects the HYDRA Machine Data Collection. The system does not correct the resource performance accounts of the currently running OP online!

Please note for data maintenance:
The tabular event maintenance of the MOC shows all changed machine statuses. However, you cannot edit the transfer posting event as it is locked. In order to perform recalculations correctly with respect to orders and machines, change the original event with the status "NOT ASSIGNED" to the correct status. The transfer posting event does not affect recalculation!

Minimum malfunction time

Specify a time in seconds for the minimum malfunction time. This value defines the time that a malfunction/disturbance must continue before the machine changes from the status "Production" to the status "Not assigned".

If operating signals are monitored, the status is directly changed. You can use the following explicit option in the MDEB2.ini to disable this behavior (deactivation of direct status change). Result: the status is only changed when the **minimum disturbance time** has expired:

MDEB2.INI

```
[INIT]
;Activating the direct status change (globally or for a specific machine)
SetMStatusDirect=1
SetMStatusDirect@<machine number>=1

;Deactivating the direct status change (globally or for a specific machine)
SetMStatusDirect=0
SetMStatusDirect@<machine number>=0
```

Minimum cycle time

If you select the **cyclic** monitoring option, specify a minimum cycle time in seconds in this field.

The terminal uses this minimum cycle time and the target cycle that is stored with the (logged in) operation and that is set off against the cycle extension to calculate the maximum value. The terminal uses this maximum value as the default cycle time.

If both, the minimum cycle time and the target cycle stored for the operation, are 0, the default cycle time is set to 60000 seconds [per 1000 machine clocks].

Cycle extension

If you select the **cyclic** monitoring option, enter the percentage for extending the target cycle time in this field. Enter a value ranging between 0 and 5000.

The system offsets the target cycle stored with the (logged in) operation against this percentage. A value less than 100 is a shortened cycle; a value greater than 100 is an extended cycle.

Number of target cycles

If you select the "**cyclic** monitoring" option, enter the number of cycles (0 to a maximum of 9) after which the terminal automatically switches from a status unequal to "production" into the "production" status within the cycle time (requirement: the status that is unequal to production is not locked for the "production" status).

Some production processes provide machine cycles during the setup phase. Set a value greater than 0 in order to prevent the current machine status from changing immediately. Please note: The quantities you collect until the machine switches to the "production" status are neither posted as yield nor scrap.

Cycles to be evaluated

Reserved Enter 0 in this field.

Management

Posting during production lock

Use this setting to specify how to post the counting pulses that are collected while the status "production" is suspended. This configuration takes effect for all counters configured as "Yield".

Posting as scrap If this option is configured for the counter, the system offsets the counting pulses against the partitioning/ pulse factor and posts these pulses as scrap. Even if you defined another quantity account for offsetting, this one will not be used.

Posting as yield parts the system posts the counting pulses as yield

No posting the system does not post the quantities while the "production" status is suspended.

Pulse factor specific to machines

Use the pulse factor, for example, if you want to collect lengths (e.g. using a wheel).

Set the value to 0 for machines where a discrete or integral number of quantities (e.g. pieces) is collected per pulse. In this case, the pulse factor is not evaluated. That means, the number of cycles posted corresponds to the actual pulses transferred via the MSS (machine interface).

The MSS (machine interface) records the signals transferred from the machine (counting pulses). According to the configured number of pulses, the system calculates and posts the quantities as follows:

Quantity for the machine = pulse * partitioning for the machine/ pulse factor for the machine

Quantity for the operation = pulse * partitioning for the operation/ pulse factor for the operation

Please note: The pulse factor will be calculated as a fraction. When the quantity is calculated, the pulse is used as denominator and the partitioning is the numerator.

The system interprets pulses that occur during a malfunction or a production lock (configuration of *Posting during prod. lock* > scrap) as scrap. Also use the above-mentioned formula to calculate the scrap quantities.

Partitioning specific to machines

Enter the partitioning specific to the machine in this field. Multiply the machine-specific partitioning by the partitioning stored with the operation in order to integrate the machine-specific partitioning into quantity calculation. Enter the value 1 in this field, if you do not want this to happen.

Extended weekend automatic

If you select this option and the system is configured accordingly, the system assigns at the beginning of the shift the status that was available before status 999 was activated.

Note:

To use this option, the workplace must already be assigned to a terminal.

Find detailed information about the automatic activation of status 999 in the document .

Waiting period, short-term disturbance

Configure a *short-term disturbance* status for each machine/ workplace to improve the overview, e.g. in the machine history. Use this status as a “repository” for unconfirmed statuses, which only existed for a specific (short) period.

If the terminal automatically identifies a downtime and the machine automatically goes back into production, the system checks if this disturbance is shorter than the time period configured for short-term disturbances.

If this is the case, the still unfounded malfunction is justified with the status that is configured as the "short-term disturbance" status for the machine.

Inputs/ outputs

Machine lock/ Target quantity reached/ Machine downtime/ Free I/O

Enter the logical output where a digital signal should occur when the corresponding status is available.

Machine lock output The system sets this output, if you enabled the option "set machine lock output" in the current machine status.

Target quantity reached output The system sets this output, if the collected yield reaches the target quantity of the OP.

Machine downtime output The system sets this output, if the machine is in a status unequal to *Production.* When changing to the *production* status, the system sets the output back to 0.

Free I/O Free input/ output for customizations.

Use these statuses for connecting a monitoring light or a horn, for example.

Enter the corresponding number in one of the fields in order to assign an output and to specify which relay is interconnected by the terminal when the predefined status occurs. Enter "0" to prevent any action. Note that you cannot assign a terminal output more than once.

Please note

Specify the statuses that trigger the activation of the machine lock in the *Status assignment*.

Generally, enter the value "1" in the input field, when the machine lock is activated via the available relay output of a DS 100. In this case, the system sets the machine lock if

- a correspondingly configured status occurs and
- the status is *not assigned*.

Output batch change**

Customer-specific assignment of an input with an automatic output batch change (MPL). By default, enter 0 in this field.

PDE (Process Data Collection)

Collect process data

This parameter specifies if the system collects process data for this machine. If this parameter is not set for a machine, you cannot collect process data for this machine.

External connection

The AIP 8.2 and/or the PCC in stand-alone mode (MDE-Blade 2 Version 8.1.0.1) do no longer support the options marked with **. As they use other configurations for the connection.

External connection

If this machine is assigned to a master terminal the following connection options are available:

No external device	External devices are not connected
DS100	DS100 connection
Arburg control system**	Arburg connection
Engel interfacing**	Connection of Engel machines
MT3**	MT3 connection
PDE**	Process data collection

If you activate a DS100 or MT3** connection, you can select the field "device address". If you activate the option "Engel interfacing", you can select the field "serial number". If you activate the option "Arburg server system", you can select the field "class".

Note regarding the combination of connections on a master terminal:
"DS 100" and "No external device": allowed
"MT 3" and "No external device": allowed
"MT3" and "DS 100" <u>not</u> allowed!

Serial number (Engel interfacing)**

Enter the serial number of the connected Engel machine. Set the option "EMS machine interface" in the HYDRA basic parameter settings if you want to use Engel machines.

Device address

You can select this field, if you activate a DS100 or MT3** connection. Enter the device address of the sub-bus participant.

"Resource configuration" tab

For resources of type "MNR", only the fields marked with "*" are available:

- Family (section *resource master data*)
- Cycles (section *target utilization*)
- Runtime (section *target utilization*)

Resource master data

Type

Identifies the type of resource:

Resource: A resource can be uniquely identified, i.e. the resource is actually present. Its quantity is always 1.

Anonymous resource: An anonymous resource cannot be uniquely identified. If the identifier is set, then you can change the value in the field *Number* from 1 to another positive integer value. You cannot post data onto anonymous resources because anonymous resources do not relate to one specific resource.

Required resource: A required resource stands for one or more actual resources that can be identified. Specify in the configuration *WRM: Master data* > *Required resources* which resources are represented by a required resource. The number results from the number of actual resources assigned to the required resource.

Please note: If this field is empty, the resource is implicitly an ("actual") resource.

Equal type

Reserved for future modifications.

Version

Revision number; store here the program version for resources of the type DNC.

Quantity

You can only edit this field, if it contains an anonymous resource and the option *Anonymous resource* is set (see above). A value > 1 indicates how many of these resources are available.

This field is calculated automatically for required resources.

Family*

Assign a resource family. If you change the resource family subsequently, an information dialog appears as a warning because user fields might possibly be assigned via the resource family.

Target utilization

Cycles*

The field *Cycles* provides additional information. The cycles value defines how long the resource is to be used.

Runtime*

The field *Runtime* provides additional information. It defines how long the resource is to be used.

Input unit

Input unit

Absolute value limit (EMG 8.1, function authorization: resablim)

Enter the absolute value limit of the (meter) resource. The energy monitor shows this limit value in addition to the current meter reading. Use the Escalation Management to generate an escalation message, if the counter value of the resource exceeds the specified absolute value limit. You need the function authorization "resablim" to view this field.

Actual utilization

The periods when a resource was logged on to a workplace are the basis for posting the *cycles (clocks), runtime, yield,* and *scrap* as *actual utilization.*

Clocks

The cycles (clocks) posted for the resource up to now.

Runtime

The total time in hours posted for the resource up to now. The total time is the sum total of all times posted onto RPA 1 to 11.

Yield (B)

The yield posted for the resource up to now (base quantity unit).

Yield (P)

The yield posted for the resource up to now (primary quantity unit).

Scrap (B)

The scrap posted for the resource up to now (base quantity unit).

Scrap (P)

The scrap posted for the resource up to now (primary quantity unit).

Configuration

Target cycle

Target duration in seconds for 1000 machine cycles if this tool is used.

Please note: The target cycle stored in the OP is relevant for the planning in the HLS module and for the machine data collection at the terminal.

Original partitioning

Partitioning of the tool (= number of cavities) when using this tool.

Current partitioning

Current partitioning of the tool. This value can deviate from the original partitioning, e.g. if the original quantity can no longer be produced with one cycle/clock due to a tool defect.

Always use the current partitioning to post cycles to the tool.

Please note: The partitioning stored in the OP is relevant for the planning in the HLS module and for the machine data collection via the terminal.

Partitioning due to cavities

If you set the option "partitioning due to cavities", the system (re-)calculates the fields "current partitioning" and "original partitioning" using the values defined in the cavity management. Then, you can no longer change the fields manually.

Log on with OP

Use this option to specify whether or not you want to log on the resource with the OP. To do so, the resource must be included as a component in the operation's list of production resources and tools. Possible values:

None: The resource is not logged on.

Implicit: The system automatically (implicitly) logs on the resource that is assigned to the operation as a production resource and tool; you can neither log on the resource manually (explicitly) nor change the logon.

Explicit: You can manually (explicitly) log on the resource that is assigned to the operation as a production resource and tool or you can log on another resource instead. If you do not log on the resource or another resource explicitly, the system implicitly (automatically) logs on the current resource; in this way, the current resource serves as a "default".

Please note:

If you log on another resource explicitly (manually), this resource will be logged on for the resource that has the same *resource type* in the operation's list of production resources and tools. For this reason, you can only log on those resources explicitly (manually) that are included as a requirement in the operation's list of production resources and tools. In this way, you cannot log on a resource that is not included as a requirement in the list of production resources and tools (the resource must be entered in the list).

In general, you should not enable this option for the resource type DNC. The DNC product group handles this differently (NC programs are logged on separately).

The system also logs on resources that are defined in the BOM of the machine.

Parallel logon/ planning possible

You can log on/plan the tool simultaneously.

Please note: You can only log on a resource to one machine more than once. Consequently, the option "Parallel logon possible" refers to several different OPs logged on to one machine.

In this case, the system posts data proportionately as follows:

- Post quantities proportionally.
- Post times 100% for each resource. This means that the system posts double the time to the resource, if the resource is logged on twice.

Post to resource

Specifies whether or not the quantities and times are posted to the resource. Due to a high degree of complexity, you should only assign this option to those resources that you actually want to evaluate.

Collective lock

If you lock a lower-level (assigned) resource using the BOM function, the system sets a collective status for the higher-level resource. If this collective status is set, the system treats the higher-level resource as locked when a download request is made.

If you enable this function, the system passes the collective lock to the higher-level resource.

Planning

Setup time

Duration in hours for setting up the tool.

Please note: The setup time stored in the OP is relevant for the planning in the HLS module.

Teardown/retooling time

Duration in hours for removing the tool.

Please note: The retooling time stored in the OP is relevant for the planning in the HLS module.

Assignment

Not used. The system uses the configuration option of the same name stored in the resource type to integrate the resource allocation in the HYDRA Shop Floor Scheduling.

Evaluation

Integrate in evaluations

Reserved for future modifications.

File

File exists

Shows whether or not the file is stored in the specified path. A cyclic process checks the files and sets the options subject to whether or not the file is available.

File name

File name; without file extension for DNC. The system adds the file extension according to the configuration in the resource type. The defined paths specify the storage location.

Comparison resources

Enter two comparison resources for energy consumption resources. They will then be shown in comparative evaluations/reports, e.g. the energy monitor.

Resource 1

Resource number of the resource to be compared.

Resource type 1

Resource type of the resource to be compared.

Resource 2

Resource number of the resource to be compared.

Resource type 2

Resource type of the resource to be compared.

Accuracy

Enter more detailed information on measuring accuracy and measuring range for test equipment resources.

Tab *User fields*

You can use user fields to store additional customer-specific information in the MES. The *user fields* tab includes eight sub-index tabs, which each has eight additional user fields. The so-called user field key specifies the available user fields and their meaning.

The workplace and resource configuration provides data of two basic object types. You can also edit this data in the workplace and resource configuration: on the one hand these are machines and workplaces and on the other these are the resources. Machines and workplaces are also "resources". But resources are not automatically machines and workplaces.

Object type

The system configures the user fields of machines/workplaces in relation to the object type "MNR". The system stores data contents to the machines/workplaces table and the resources table of the database to ensure data consistency.

The system configures user fields for resources in relation to the object type matching the resource type of the resource (example: create resources of the type "PAC" in relation to the object type "PAC"). The system stores data contents to the the resources table of the database.

User field key

Each user field key describes a combination of user fields. The management of the user field key (and therefore the meaning of the fields) is different for each object.

User fields

The following user fields are available after configuration:

Field data type	Number of fields
Date	6
Numeric, time, duration	16
Decimal value	6
Text field, length 1	16

Field data type	Number of fields
Text field, length 10	6
Text field, length 20	14
Text field, length 40	2

Each page shows a maximum of 8 fields.

By default, no user field keys are defined. Configure the system accordingly to support this kind of user fields.

As the table shows resources of different types, use the user field key "SYSTEM" of the object "RES" to identify the column headings for the user fields.

Comment tab

Store additional resource comments in the "comment" tab.

Main tab *Resource attributes*

Shows additional resource attributes via the user field definitions of the resource family. Use the "resource attributes" button for editing.

Main tab *Resource list*

Shows the resource list for the selected resource. Click the "resource list" button to go directly to the BOM application for editing purposes.

Main tab *DNC versions* (available as of DNC 8.2)

Shows the available versions of a DNC resource including a flag indicating the currently applicable version. HYDRA provides this valid version for machine downloads.

Toolbar

General tab

Insert

Function authorization: mdres.create

Opens the dialog for adding a resource. This dialog provides the fields that match the selected resource type.

Copy

Function authorization: mdres.copy

Opens the dialog for copying an existing resource. Subject to the selected resource and its resource type, the copy function differentiates the following:

- Copy function for resources of **resource type = MNR** (workplaces, machines)
- Copy function for resources that do **not** have the **type MNR**

Copy function for resources of resource type = MNR (workplaces, machines)

From: resource type, resource, short name, name

- Resource type (fixed "MNR“)
- Workplace/machine number
- Short name
- Name

of the workplace you want to copy. You cannot change these values. They derive from the selected data record.

To: resource type, resource, short name, name

- Resource type (corresponds to the resource type of the workplace you want to copy; cannot be changed).
- Workplace/machine number
- Short name
- Name

of the target workplace.

Copy machine status

Function authorization: mdmst.copy

If you set this option, the system automatically creates and transfers all of the workplace you want to copy to the new workplace.

Copy counter configuration

Function authorization: mdctr.copy

If you set this option, the system automatically creates and transfers all of the workplace you want to copy to the new workplace.

Note that the counter numbers of the new workplace are identical with the counter numbers of the workplace you copied. If necessary, you have to adjust the counter numbers.

Copy reasons

Function authorization: mdreas.copy

If you set this option, the system automatically creates and transfers all of the workplace you want to copy to the new workplace.

Copy function for resources that do not have the resource type MNR

The copy function for all resources that do not have the type MNR opens the "insert" dialog and takes over the details from the previously selected resource. But you can edit and change all fields.

Edit

Function authorization: mdres.edit

Opens the dialog to edit a resource and provides the tabs and fields of the relevant resource type.

As of MES Weaver 4.0$_{pe}$, you can change master data of several selected resources of the same resource type at the same time. You can select up to 10 fields and assign a value. You require the function authorization ***mdresmm*** to edit several resources at once.

Delete

Function authorization: mdres.delete

Deletes one or several selected resources.

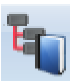
Resource tab

Configuration – resource status

Opens the application "resource status" to define statuses for all resources that do not have the type MNR.

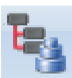
File - show file

Opens the file view – only available for document resources, which are configured as file-based resources without DNC processing in the *Resource type.* And only available if the relevant license and function authorization are available.



Go to - resource list

Opens the *Resource list* application. The selected resource is entered as default value for the higher-level resource.



Go to – required resources

Opens the "required resources" application. The selected resource is entered as default value for the required resource.


Go to – cavity assignment

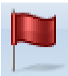
Opens the "cavity assignment" application. The selected resource is entered as default value.


Go to - resource attributes

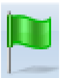
Opens the application "resource attributes". The selected resource is entered as default value.

Functions – Measures

Opens the *Measures* application.


Functions – Status change

Opens the dialog to change a resource status. The checkbox *Including subordinate resources* is not relevant and reserved for future extensions.


Functions – Release of resource

Opens the dialog to release a resource. The checkbox *Including subordinate resources* is not relevant and reserved for future extensions.


Functions – Stock transfer

Opens the dialog to transfer/relocate a resource.

Workplace tab


Configuration – status assignment

Opens the application "status assignment". The system enters the selected resource in the corresponding field.


Configuration – counter configuration

Opens the application "counter configuration". The system enters the selected resource in the corresponding field.


Configuration – terminal assignment

Opens the application "terminal assignment". The system enters the selected resource in the corresponding field.

 **Entry – reasons**

Opens the application "reasons". The system enters the selected resource in the corresponding field.

 **Entry – Operator positions**

Opens the application "operator positions". The system enters the selected resource in the corresponding field.

 **Entry – premium indicator**

Opens the application "premium indicator". The system enters the selected resource in the corresponding field.

 **Groups - groups**

Opens the application "groups". The system enters the group of the selected resource.

 **Groups – group assignment**

Opens the application "group assignment". The system enters the selected resource in the corresponding field.

 **Miscellaneous – cycle parameter**

Opens the application "cycle parameter". The system enters the selected resource in the corresponding field.

 **Miscellaneous - workforce requirements of workplaces**

Opens the application "workforce requirements of workplaces". The system enters the selected resource in the corresponding field.

DNC tab

The tab is only available, if you select a DNC resource. These are resources configured as resources with DNC processing in the resource type.

 **Configuration – resource status**

Opens the "resource status" application.

Configuration - assignment of DNC family to machine

Opens the application "assignment of DNC family to machine".

Copy resource attributes (as of DNC 8.2)

Copies values of resources attributes from one resource to another. Both resources must use the same user field key.

File - comparison editor

Opens the comparison editor for the selected resource or resources. See below for further information.

File - export

Exports the file specified for the resource. You use the file explorer to specify the target file.

File - import

Imports the file specified for the resource. You use the file explorer to specify the source file.

File - viewer

Opens the file specified for the resource using the defined viewer program.

File - editor

Opens the file specified for the resource for editing using the defined editing program.

Set valid version (as of DNC 8.2)

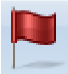
Only active, if you select a version in the *DNC versions* tab. The selected version is set as the new and valid version.

Go to - resource attributes

Opens the application "resource attributes". The selected resource is entered as default value.

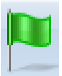
Go to - resource list

Opens the *Resource list* application. The selected resource is entered as default value for the higher-level resource.



Functions – Status change

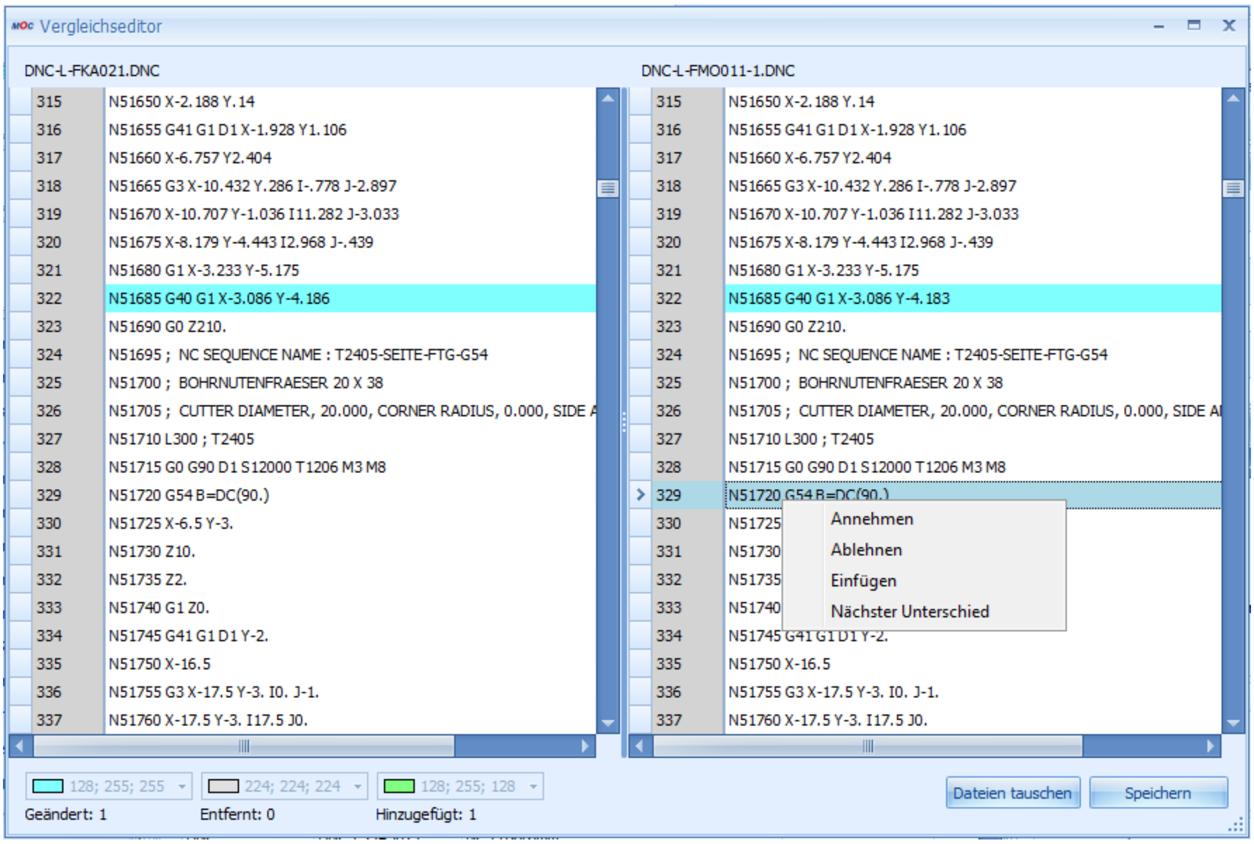
Opens the dialog to change a resource status.



Functions – Release of resource

Opens the dialog to release a resource.

How to use the comparison editor



The comparison editor compares the files attached to the DNC resources. Two operation modes are available:

Selection of one resource:

> The editor shows the released resource and the optimized version of the resource for comparison. You can change the file displayed on the right-hand side of the editor. Once you have made the changes, the comparison editor transfers these changes to the system, like the simple editor. You can only use this mode for DNC types with the file processing type "optimized".

Selection of two resources:

If you select two resources before you open the comparison editor, the editor compares the two selected resources. You can select the file type. You can change the file displayed on the right-hand side of the editor. Once you have made the changes, the comparison editor transfers these changes to the system, like the simple editor.

Click the relevant buttons or use the context menu (right clicking) to start the functions of the comparison editor:

- Reject: Rejects the difference identified (on the right). Accepts the value from the left file. The editor does no longer highlight the difference.
- Keep: Accepts the difference identified (on the right). The editor does no longer highlight the difference.
- Next difference: Goes to the next difference.
- Insert: Inserts a row at the current position.
- You can always change the contents of a row. Click the row and enter a value. Press ESC to quit the row without changes. The editor then highlights the row as "changed".
- Swap windows: Click this button to swap the windows. This function is necessary if you compare two resources. The place where a resource is displayed results from the display order in the table; the system does not know, which resource must be changed. If you only select one resource, this button is not available as in this case you can only change the optimized program version.
- Save: Saves the changes made to the file on the left-hand side.

Processing notes for workplaces and machines

Configuration changes

Restart the terminal which the workplace/machine is assigned to in order for the terminal program to interpret the configurations or modifications made to this workplace/machine.

Deleting a machine/ workplace

In a first step, the system shows a confirmation prompt asking if you really want to delete the machine. If you confirm this prompt, the system makes an attempt to delete the workplace. You can only delete a workplace successfully, if:

- you have not yet collected data for the workplace;
- you have currently not assigned the workplace to a terminal or a line;
- you have currently not logged on operations to the workplace;
- you have not planned operations for the workplace.

If you delete the workplace successfully, the system also deletes all configuration data, e.g. status assignments, for this workplace.

Checking Business Parameter Containers (BSCs)

See for further details on how to check the system against business parameters.

8 Status Texts

Overview

FEDRA menu	Master data → Workplaces/ machines → Status texts
FEDRA menu	Master data → Workplaces/ machines → Status texts
Transaction code	Mstt
Function authorization	Mdmstt

In the status text dialog, all possible states (statuses) are designated with a descriptive text. These descriptions are then used in the status assignment performed later. The objective is to use standard status texts for all workplaces and machines.

Purpose

In the status text dialog, all possible states (statuses) are designated with a descriptive text. These descriptions are then used in the status assignment performed later. The objective is to use standard status texts for all workplaces and machines.

Integration

During status assignment, a status text is assigned to a status.

The colors configured in the status text can be displayed in different evaluations/reports.

Selection criteria

The application provides the following selection criteria:

Status text

Unique status text number

Designation

Status text name

Field descriptions

Status text

Unique status text number

Designation

Informative description of the status. This text is displayed at the terminal, among other places.

Color

You also have the option to assign a color to a status text. This color is taken into account at various places in MOC.

Recommendation:

When setting up a new status text, the status number should preferably match the number of the status that will later be created during status assignment.

Example: The status text 2 "Set up" should be assigned to the status 2 during status assignment.

9 Status Assignment

HYDRA menu	Master data → Workplaces/ machines → Status assignment
FEDRA menu	Detailed scheduling → Master data → Status assignment
Transaction code	mst
Function authorization	mdmst

You can create statuses for all workplaces/ machines. The status shows the current status of the workplace/ machine.

Purpose

All possible statuses or malfunctions at the machine/workplace are configured in the *Status assignment* and assigned to the status texts by distinct status numbers. If malfunctions are detected on the terminal, the system uses the statuses documented in the status assignment.

Example of how a status table is set up :

Status	1	2	3	4	5	6	12
Status text number	1	3	4	5	6	2	12
RPA	MUT	LCI	SCI	SET	BKS	DCI	DCI
Control indicator	Production	Malfunction	Malfunction	Malfunction	Malfunction	Malfunction	Gen. disturbance
Manual assignment via the terminal (Manually at terminal)	☐	☐	☐	☑	☑	☐	☑
Automatic assignment via digital input	☑	☑	☑	☑	☐	☑	☐
Digital input	1	2	3	4	0	6	0
Assignment on the machine interface	0	2	1	3	-	7	-

The entries are based on the status texts listed below:

1 Production

2 Tool breakage

3 No raw material

4 Staff shortage

5 Setup

6 Break

12 Gen. disturbance

This is only an example illustrating the status assignment.

The selection dialog allows you to select and view the statuses that have already been assigned to a machine or a workplace.

Integration

The workplace/ machine status is integrated in various evaluations/reports.

Requirements

You have to create the following objects before you can assign workplace/ machine statuses.

- Workplace/machine
- Status text
- Status class (optional)
- Resource Performance Account (RPA)

Toolbar

Status list

Click the "Status list" icon to open the report including the defined and selected statuses. This report shows the statuses in printable form as plain text and as bar code. Click here Machine status list report to find detailed information.

Selection criteria

The application provides the following selection criteria:

Workplace

Select the workplace/machine for which you want to display the statuses.

Status

Select the defined workplace/machine statuses. The application shows the statuses of all workplaces/machines matching the entered status number.

Status text

Select the defined status text. You can also use wildcards.

RPA abbrev.

Abbreviation for the assigned Resource Performance Account

Status class

Abbreviation for the assigned status class

Field descriptions

Notes on status 30000

You can only change selected options for status 30000:

- Warning in the graphic machinery
- Activate production lock
- Activate machine lock
- Record scrap reasons

Certain fields such as status class, RPA, control indicator are not filled with this status and are therefore empty in the list.

General tab

Machine/workplace (short name)

Enter the machine number to assign the disturbance status to a machine. Enter the same number in this field for all statuses of a machine in order to create a complete status table for a machine.

The *short name* includes the name of the entered or selected machine.

Status

This field includes the unique number for statuses included in the status table. You can also use this number to assign and/or change the status via the terminal.

You can only define one status for workplaces of the type "group workplace". Assign the characteristic "production" (see "control" tab) to this status.

Note: You cannot delete the status 30000 "Not assigned". You can only configure specific fields for this status.

Superior status

If no status is defined in the Superior status field, the currently created status is at the highest level.

Otherwise, enter the number of the directly superior status. This must already exist and have the control indicator *Hierarchy level*.

Note: This function is available only for Windows terminals.

Status text

The number assigned here refers to the plain text status message from the status text table.

Status class

Assign a status to a status class to make cumulative evaluations/reports about status classes. The abbreviation assigned here refers to the plain text status class message from the status class table.

Resource Performance Account (RPA)

Enter a value in this field to assign the status to a Resource Performance Account (RPA). Select one of the 12 Resource Performance Accounts (RPA).

By default, the 12 Resource Performance Accounts are already defined in HYDRA. Refer to the glossary for further information on the Resource Performance Accounts.

Control tab

Control indicator

The following characteristics are available to specify machine monitoring.

Except for the "production" characteristic, all following characteristics are only allowed for machines/workplaces of the type "individual workplace".

Production

Production identifies the production state/status of a machine/workplace. Assign the "production" characteristic to **exactly one** status for each workplace.

If the machine monitoring system detects production signals, then the status of machines/workplaces of the type "single workplace" is changed to the status to which this control indicator is assigned.

Only one status is allowed for workplaces of the type "group workplace" and this status must be assigned this control indicator.

Other status

Assign the control indicator "other status" for all statuses without a control indicator. You can assign this control indicator/characteristic to any number of statuses for each individual workplace.

General disturbance

Create exactly one status as general disturbance for each machine/workplace. If the machine data collection detects a production phase that has not yet been assigned a disturbance or status, then this duration is posted to the status assigned to the control indicator for the general disturbance.

Material change

This option is only available with the MPL module.

If a workplace has a status with this indicator, materials that are not planned can also be logged on. These materials are logged on as alternative material, which means that you log on some other material instead of the planned material. You can specify in the status assignment if alternative materials may be logged on. As a result, you have to configure at least one "Setup" status with the "Material change" option in order to enable the desired posting behavior. If the machine is in a status that is assigned the "material change" option, you can log on the alternative material already when you log on the operation. In this case, you do not even have to change the status beforehand.

No order

If you interrupt or log off an operation manually, the Windows terminal verifies whether this is the last operation of the workplace. If no more operations are logged on to this workplace, the terminal sets the workplace status to the status assigned to the "No order" option.

This option is only available for Windows terminals (CTWIN/AIP).

A status with this control indicator may only be configured at machine/workplace of the type "single workplace". You may configure only one status with this control indicator.

Do not configure a status with the control indicator "No order" at workplaces of the type "group workplace", since only the status "Production" is available at group workplaces.

Short-term status (as of MDE 7.2)

For an optimized overview, for example in the status log or machine history, you can configure one status per machine as a short-term disturbance. Use this status as a “repository” for unconfirmed statuses, which only existed for a specific (short) period.

If the terminal identifies a downtime and the machine automatically returns to the status "production", the system verifies if this disturbance took less time than configured for short-term disturbances.

If this is the case, the still unfounded malfunction is justified with the status that is configured as the "short-term disturbance" status for the machine.

The system does not differentiate between such automatic status postings (automatic assignment of reasons) for short-term disturbances and reasons entered manually by operators. The duration of short-term disturbances is ignored with automatic shift changes.

Hierarchy level

Assign the option "hierarchy level" to statuses that cannot be recorded via the terminal. These statuses only represent the hierarchy.

This function is only available for Windows terminals. In this case, you can neither change specific configurations nor enter data in certain tabs.

Estimated downtime

If you assign a status manually, you can enter an estimated downtime. In this case, the application suggests the downtime that is stored in the master data.

If statuses are changed automatically, the system assigns the downtime stored in the master data automatically.

Activate production lock

If this option is set, the production lock (P lock) is automatically activated when a status is assigned via the terminal.

If you use the machine monitoring function, the *production lock* option prevents the machine from switching automatically to the status "production" when a production signal arrives. Consequently, this status overrides the production signal until you manually disable the production lock option. The production lock can also be used to determine whether and how quantities (counter readings) are posted during this time.

Setting the machine lock output

You have to set this option, if you want the machine lock to be enabled when assigning statuses.

In this case, you also have to ensure that the machine lock output has been configured accordingly in the machine configuration.

Warning in the graphic machinery

The entry determines the time after which the symbol (more precisely: the part of a symbol that represents the status in color) starts flashing in the Graphic machinery after the workplace/machine status has occurred. Enter the value in the format hours: minutes: seconds.

Status change tab

Manually via the terminal

If this option is selected, you can enter the status manually via the terminal (using barcode or keyboard). If this option is not selected, the status selection list of the terminal will no longer show the status.

Authorization

Access authorization for entering a status via the terminal (enter a value between 0 and 9). An authorization level for machine status modification is defined for every person in the HR master data. If the authorization level stored in the master data is lower than the authorization level defined here, you cannot assign the status via the terminal.

Automatically via digital input

Select this option, if you want the statuses to be assigned automatically via the machine interface (CT-MSS, CT-UMPS, PCC). Enter the number of the digital input identifying the status in the "digital input" field (0 = no input).

If you monitor machines via the operating signal , a digital input also records the operating signal. Proceed as follows to define a status as an operating signal:
- Control for machine monitoring: "Production"
- Select the option "Automatically via digital input"
- Enter a value > 0 in the "digital input" field. This input records the operating signal.

In the case of disturbance reasons, a general distinction is made between automatically and manually recorded disturbance reasons. Distinguishing features are:

- Disturbance reasons you enter manually at the terminal override automatically set disturbance reasons.
- Operating signals do not override a status you set manually. Except for operating signals for the status "production" (see next bullet point). The "production" status also overrides a manually set status.

If no production lock is set, the status with the control indicator "Production" overrides every disturbance reason. Therefore, keep in mind that the status "Production" must be deactivated, if you want a current disturbance to be processed at the input.

If multiple automatic statuses (disturbance reasons) are recorded via digital inputs, the status with the lowest HYDRA channel number (not the lowest status number!) is set.

Note that the assignment of the number to the physical connection at the MSS depends on the settings in the local terminal configuration file (Windows terminal: CTWIN.INI/CTAIP.INI, DOS terminal: AIOP.CFG).

If a status is to be processed automatically by the machine based on the transferred HYDRA status number (MSTAT), this option "Automatically via digital input" must be activated and the field "Digital input" must be assigned the value "0".

If a digital input and a status number (MSTAT) for a machine status are set at the same time for a machine, the status of the digital input at the machine is set. The digital input therefore has a higher priority than the status via MSTAT.

Digital input

Number of the digital input used to set the status.

Status transfer to aggregates

This option allows you to set a global status at the production line, which is then automatically set for all aggregates assigned to it.

The requirement for this is that this status is also configured for all assigned aggregates and that the status number for the aggregates is identical to the status number of the production line.

Processing tab

Log off staff

Please note: the weekend automatic function (status 999) does not support this option.

If you select this option, the system logs off all persons currently logged on when a status is assigned (useful during maintenance phases). Otherwise, the persons stay logged on.

Operation posting

Please note: the weekend automatic function (status 999) does not support this option.

Use the option "operation posting" to have an overhead cost order logged on automatically when statuses change. The following options are supported:

None

No processing.

Interrupt OP

Use this setting to interrupt automatically all active operations and to log off all employees from the workplace if this machine status is set.

Interrupt active OP and log on the following OP

Use this option to interrupt all active operations and to log off all employees when statuses are changed. The system automatically logs on the "subsequent operation" stored in the *Operation* field.

Please note: The subsequent operation must not be subject to batch management.

Transfer registered persons to OP

Depending on this setting, the system transfers the persons logged in to the "subsequent operation" defined in the *Operation* field.

If the operation is an overhead cost operation or if the workplace is a group workplace (GAP), then at least one person is logged on to the subsequent OP:
- either the person carrying out the posting, or
- the person who is logged on the longest if the *change status* dialog does not include a field to enter the staff badge number. If no employee is logged on at this time, the subsequent operation cannot be logged on.

Scrap reason

Depending on the current status, you can post automatically recorded scrap to a defined reason.

A distinction is made in the process, whether the production lock is set or not while the status is active. You can choose from the following configuration options:
- scrap reason
- scrap reason during production lock

A counting input explicitly defined as a scrap counter generally takes priority over a reason defined here.

Plausibilities tab

Check running operation

You can configure statuses,

- if an OP is logged on to the workplace	Use Option *An operation must be logged on*
- if no OP is logged on to the workplace	Use Option *An operation must not be logged on*
- if an operation is logged on to the workplace or not	Use Option *No check*

HYDRA checks this dependency during the manual status change and during the manual login/logout or interruption of operations. If the posting violates the condition, the terminal issues an error message and refuses the posting.

If a new status is set when an operation is terminated or interrupted, then this status cannot be an order-related status!

Use of unplanned material allowed

This option is relevant if you use the HYDRA module Material and Production Logistics (MPL). If the option is set, you can use unplanned material. That means, you can log on batches that are not specified in the input material list of the operation as input material. This can be useful during setup.

Tab *User fields*

User field key

This field of the object type MST is preset with the user field key DEFAULT. Normally, you cannot change this assignment. MPDV defines the user fields for this user field key during the customizing process to meet specific customer requirements.

10 Setting Outputs Depending on Status and Posting Scenario

Usage

Under advanced configurations, it is possible to set the machine's digital outputs to account for when a certain machine status and a defined posting scenario coincide.

The following combination of machine status and posting scenario can be configured:

- A status is active without a production lock
- A status is active with a production lock
- A status is active, but no operation is logged on
- A status is active and at least one operation is logged on
- A status is active, but no person is logged on
- A status is active and at least one person is logged on

If the situation changes (status modification, activate or deactivate P lock, OP posting, personnel posting), the terminal checks the logical configured outputs and sets them as necessary.

In doing so, there is the option to set exactly the same logical output for several statuses within one machine as well as several statuses of different machines.

If a logical output is set for different conditions at the same status, then it will be set if one of the conditions is met (OR link).

Requirements

- This function is only available at the Windows CTWIN or AIP terminals.
- For this purpose, the terminal must be configured in operation mode "MDE".
- The machine must be permanently assigned to the "MDE terminal".
- Output signals that are used for this function may not be configured anywhere else at the same time (e.g. machine lock). This must be assured during configuration. This is not checked automatically.
- The terminal must be restarted any time a change is made to the configuration of the logical outputs.
- The outputs configured here may not be used simultaneously with the "inputs/ outputs" configuration that exists in the resource configurations.

Procedure

The configurations required for this are defined in the so-called "object-related configuration".

Object type fix "MSTAT" (Machine STATus)

ID1 Machine number

for a numeric machine number, always add leading zeros to make it eight digits

ID2 Machine status number

ID3 (leave empty)

ID4 (leave empty)

Parameters The following parameters are supported:

MST_NO_PLOCK	Output/ signal for "Status active without production lock"
MST_PLOCK	Output/ signal for "Status active with production lock"
MST_NO_ANR	Output/ signal for "A status is active, but no operation is logged on"
MST_WITH_ANR	Output/ signal for "A status is active and (at least) one operation is logged on"
MST_NO_PNR	Output/ signal for "A status is active, but no person is logged on"
MST_WITH_PNR	Output/ signal for "A status is active and (at least) one person is logged on"

Parameter value The parameter value is always the number of the logical output that is to be set.

Active Fix "J"

Example

Output 1 is to be set at machine 100 if either setup (status 2) or dismantling (status 9) is active with a production lock.

Output 2 is to be set at machine 100 if production (status 1) is active although no operation is logged on.

Object type	ID 1	ID 2	ID 3	ID 4	Parameters	Parameter value	Active
MSTAT	100	2			MST_PLOCK	1	Y
MSTAT	100	9			MST_PLOCK	1	Y
MSTAT	100	1			MST_NO_ANR	2	Y

11 Print Status List

Summary

Menu	Master data → Workplaces/ machine status assignment → Status list
Transaction code	- (mst for machines/ workplace status assignment)
Function authorization	mdmst.print

Usage

The purpose of this function is to generate a status list with clear text and barcodes. A barcode provides the ability to enter the status at a terminal using a barcode reader (e.g. a scanner gun).

Integration

The function is called up from the status assignment of machines/ workplaces.

A printed barcode provides the ability to enter the status at a terminal using a barcode reader (e.g. a scanner gun).

To print a status list, follow the steps listed below:

1. In the list, highlight the status you would like to print out on the status list. Click on the top left-side corner of the table if you want to highlight all statuses.
2. Click on the "status list" icon to call up the print preview screen. In it, there will be one new page for each workplace/ machine.
3. Click on the icon "Print report"  in the print preview to print out the status lists on the default printer set in MOC.

Structure of the workplace/ machine status barcode

NNNNN0

Place	Designation	Length
*	Asterisk	1
N	Machine status, with preceding zeros	5
0	fixed: 0	1
*	Asterisk	1
Length of status barcode without asterisks		6

Please note:

By default, HYDRA supports the barcodes "39", "128" and "Interleaved 2 of 5"

The system only supports barcode detection and automatic assignment to the corresponding input fields at the terminal for barcode readers that are connected at the serial port (COM port). This is not possible for barcode readers that are "looped in" through the keyboard.

12 Cycle Parameters

Summary

Menu	Master data → Workplaces/ machines → Cycle parameters
Transaction code	cycpa
Function authorization	mdcycl

Usage

HYDRA provides the ability to monitor cycle times within the machine data recording function without requiring HYDRA process data processing to be used.
The purpose of this function is to configure the action and tolerance limits.

Integration

Values are marked in different colors in the *Workplace overview* depending on whether the action or tolerance limit was exceeded:

Standard	Black
If the value drops below or exceeds the action limit	Blue
If the value drops below or exceeds the tolerance limit	Red

If a limit is exceeded, no further processing steps are taken in HYRDRA.

Requirement

Before defining any configurations, you must first set up the machine.

Selection criteria

The application provides the following selection criteria:

Machine

Selection by machine/ workplace

Field descriptions

Machine

Machine for which the configuration applies.

Tolerance limit positive, negative

Values may not drop below or exceed the percentage values defined here. The cycle time of the logged on operation is always used as the target value for cycle time monitoring. This can be corrected at the terminal. The limit value is entered as a percentage of the target value.

Example:	Target value:	20 sec/ cycle
	tolerance positive:	10 %
	tolerance negative:	5 %

Thus, this results in the following limit values:

	Upper limit value:	22 sec/ cycle
	lower limit value:	19 sec/ cycle

Action limit positive, negative

Percentage values can be entered here, triggering a warning once they have been reached. This is why the action limits should be defined more narrowly than the tolerance limits. The limit value is entered as a percentage of the target value.

13 Machine Counter Configuration

Overview

	→	
	→	

Menu	Master data → Workplaces/machines → Counter configuration
Transaction code	ctr
Function authorization	mdctr

Purpose

The counter configuration defines the behavior of how the pulses transmitted by MSS (machine interface) should be interpret in terms of monitoring and quantity.

Integration

In addition to defining counter configuration, each time a machine is connected to the terminal several different settings must be configured.

Requirements

Before defining any configurations, you must first set up the Machine.

Configuration changes

In order for the shop floor terminal program to be able to interpret the settings or changes that were made, you must first restart the terminal that is assigned to the workplace/ machine.

Selection criteria

The application provides the following selection criteria:

Machine

The ability to select the defined counter for the chosen resource/ machine.

Field descriptions

Machine

By entering the machine number, a logical counter channel is assigned to a machine.

Numerator

Counter number used to uniquely identify the counter channel for the machine.

Note that the assignment of the counter number to the physical connection to the CT-MSS, CT-UMPS or OPC variable depends on further settings (e.g. local terminal configuration file CTWIN.INI/CTAIP.INI (Windows terminal) or AIOP.CFG (DOS terminal).

Designation (name)

Explanation of the counter, e.g. *Yield at closure counter*

Unit

Quantity unit that is recorded by the counter. Is currently only used for documentation purposes.

Cycle monitoring

If this identifier is set for a channel, incoming counter pulses are considered during cycle monitoring.

When calculating the actual cycle, only the first counter is referenced that is configured by the identifier "For monitoring". This counter must also be set for monitoring via operating signal if an actual cycle is to be calculated at the same time.

Posting as

Defines which quality is recorded at the counter quantity. Different values:

- yield
- scrap
- rework
- open quantity

The system posts generally to the primary quantity account.

Counters that are configured with the *No posting* option do not post anything to the quantity account. Only the counter reading is collected. Cycles or strokes may be posted and/or controlled via the cycle monitoring.

Reason

If you post quantities for scrap, rework or open quantities, you can store a reason in the counter. To do so, you must first create the reason in the configuration accordingly (*Reasons*).

Thus, for example, you can define two separate counters with different scrap reasons.

A scrap reason stored in the counterconfiguration has a higher priority than a scrap reason stored in the status.

Posting as cycles

If this option has been set, the pulses coming in via this counter are posted as cycles.

Posting with partitioning/ pulse factor

If this option is set, the incoming pulses are calculated using the respective active partitioning/ pulse factor.

For counters with the option *No posting*, the setting *Posting with partitioning/ pulse factor* is irrelevant, because no cycles are converted into quantities for these counters.

Offset against

You can offset the pulses recorded with this counter and the resulting quantities against a different quantity account. In this case, the entered quantity is deducted from the specified account.

Note: If you offset quantities, the resulting values can be negative values.

What you must be aware of for DOS based terminals is that you must make sure you do NOT set the option "Offset against" in the machine/ workplace configuration, index tab "Quantities", but instead you must set a compensation here.

Counters with the option "No posting" cannot be offset against quantity accounts.

Offset against material

This identifier is used to enter the counter that collects the material consumption of the input material. Which material is consumed from the BOM can be specified using either the material type or BOM item options.

Consumption recording

You can collect the material use in different ways. You can find a general overview here Consumption recording .

Material type

If you enter a material type here, the counter is used for exactly this material of the BOM with this material type.

BOM items

If a BOM item is stored here, the counter is only used for this material.

Modified on

Modified by, date and modified on

Notes for DOS based terminals

The counter configuration is only supported to a limited extent for DOS based terminals (CT-56x, CT-541):

Only a maximum of one counter can be configured for each quantity account *Yield* or for each quantity account *Scrap* (no rework, no open quantity, and no counter with the option "No posting").

An *Offset against* is only possible at the second counter that must be configured as *Scrap*. In this case, the compensation is only possible with *Yield*.

What you must be aware of is that you must make sure you do NOT set the option *Offset against* in the machine/ workplace configuration, index tab *Quantities*, if a compensation is set here.

No reason may be defined for the scrap counter.

Toolbar

Workplace configuration

Click this button to call the Workplace and resource configuration.

14 Workplace terminal assignment

Overview

	→	

Menu	Master data => Workplaces/machines => Terminal assignment
Transaction code	wta
Function authorization	mdwta

Purpose

This function is used to configure the machine assignment for the terminal.

Purpose

The assignment of a machine to a terminal is a requirement to use MDE-specific functions such as automatic shift change, cyclic status updates or recording via MSS. However, these functions are only available if the terminal is configured as a so-called "MDE terminal" (see Terminal Configuration). Terminals configured as ADE terminals do not provide these function even if the machines/workplaces are not assigned to the terminal.

The assignment ensures that the machine/OP is displayed by default on the terminal.

The number of machines that can be assigned depends on the terminal type.

Terminal type CT-541:	only one assignment possible
Terminal type CT-76x, CT 83x, CT84x, CT850 (AIP 8.1 and 8.2):	up to 16 assignments possible (even though the terminal is configured as master terminal) up to 10 machines may be used for the process data collection (HYDRA-PDV).
Terminal type CT-56x:	up to 8 assignments possible
Terminal type A-SUB	up to 20 assignments possible.

All machines previously assigned to terminals are shown according to the selected terminals. The order in which the display is shown on the terminal is determined by the position specified here.

If a production line (only available if MDE-LIN license is available) is assigned to the terminal, all aggregates of the production line are automatically assigned to the terminal and displayed in gray under the position “99”. Aggregates cannot be removed from the assignment. If a production line is removed from the assignment, the aggregates assigned to the production line are automatically removed. Production lines can only be attributed to the terminal types CT76x, CT83x (max. 2 lines) and CT84x (max. 3 lines).

The option "Processing" can perform different assignments of the machine type to the terminal. The following options are available:

A - BDE processing

M – MDE processing

Processing as per operation mode of the terminal

Therefore, workplaces/machines with HYDRA-MDE-processing and workplaces only with HYDRA-ADE-processing may be assigned to an MDE terminal.

You have to set the processing to "BDE processing" for a group workplace, if you want to assign the group workplace to a terminal.

Number of terminal assignments

You can only assign a machine/workplace to a single terminal. If you want to assign a machine to several terminals, then you have to obtain the appropriate license.

Configuration changes

Restart the terminal to ensure that the settings or changes made can be interpreted by the terminal shop floor program.

Selection criteria

The application provides the following selection criteria:

From - to

Select terminal number

Field descriptions

Terminal

Assign a machine to a unique terminal number.

Position

Display position of the machine at the terminal and for the terminal assignment at the client.

Machine

Machine you assign to the terminal.

Processing

Here, you have the following 3 options:

A	BDE processing
M	MDE processing
	Processing as per operation mode of the terminal

15 Postings Relating to Workplaces/Machines

Summary

Menu	Data collection --> Corrections --> Postings relating to machines
Transaction code	mboo
Function authorization	mboo

Utilization

This application enables the user to obtain an overview of the postings relating to workplaces/machines generated in HYDRA and to edit, record and correct relevant data quickly and easily.

The below-mentioned documents provide further information on postings relating to machines:

- Posting of quantities
- Posting of Times

Please note: The terms "posting" and "log record" are used synonymously in this document.

Integration

A log record always refers to a period of time in which a workplace/machine status is available. Each status change results in a new log record.

If the shift automatic function is enabled, the shift change also leads to a new log record, even if the status does not change in this case. If the status 30000 "not assigned" is available when shifts change, this period is posted onto the status with the control indicator "general disturbance".

Prerequisite

Machine status changes need to be recorded for the workplaces/machines.

Selection criteria

The application provides the following selection criteria:

Workplace ... to ...

Determines the machine-related postings that were posted onto the specified workplace/machine.

Group ... to ...

Determines the machine-related postings that were booked onto workplaces/machines of the entered group (according to the resource configuration).

Cost center

Determines the machine-related postings that were booked onto workplaces/machines of the entered cost center (according to the resource configuration).

Company

Determines the machine-related postings that were posted onto workplaces/machines of the entered company (according to the resource configuration).

Responsibility area

Determines the machine-related postings that were booked onto workplaces/machines of the entered responsibility area (according to the resource configuration).

Date from ... to ...

Determines machine-related postings the start time of which is within the specified period of time. In this context, the period is either defined in connection with specifying the shift or the time.

Please note that this application only allows for data of the online data area to be selected and edited.

Shift

Restricts the date range defined above by selecting by the shift. Example: if the date range is selected from 23 January 2009 until 25 January 2009 and shift 1 is chosen, all machine-related postings the start time of which is in shift 1 of 23 January 2009, shift 1 of 24 January 2009 or shift 1 of 25 January 2009 are selected.

Time

Restricts the date range defined above by selecting by the time. Example: if the date range is selected from 23 January 2009 until 25 January 2009 and the time from 3 pm until 11 pm is chosen, all machine-related postings the start time of which is between 3 pm on 23 January and 11 pm of 25 January 2009 are selected..

If several selection criteria are used, the results matching these criteria will be displayed.

Field Descriptions

“General” tab

Record type

P: This log record was generated by a status change

N: This log record was generated at the end of a shift

Confirmed/uploaded

This flag cannot be changed. It is set within the scope of customer-specific uploads.

Machine

Machine/workplace for which the log record was generated.

Cost center

Cost center of the machine/workplace according to the workplace configuration for which the log record was generated.

Posting start, posting end

Period of time (beginning, end) for which the log record applies.

Duration

Duration since the last status change

Shift

Shift (number) for which the log record applies.

Beginning of shift

Beginning of the shift for which the log record applies.

End of shift

End time of the shift for which the log record applies.

Status

Status which coincided with the period of time of the log record.

The resource performance account is determined automatically from status assignment.

Comment

Comment (max. 60 characters) for the machine status change, if entered while assigning statuses (dialog M_MST) on the terminal.

Target cycle

Target cycle that was set when the status was changed (end of log record).

Partitioning

Partitioning that was set when the status was changed (end of log record).

Editor

Editor who modified the log record manually the last time.

Last modification

Point in time when the log record was edited manually the last time.

“Quantities“ tab

Yield (P)

Computed yield in primary quantity unit for counter values that were recorded during the period of the log record or yield in primary quantity unit that was manually recorded during the period of the log record.

Scrap (P)

Computed scrap in primary quantity unit for counter values that were recorded during the period of the log record or scrap in primary quantity unit that was manually recorded during the period of the log record.

Rework (P)

Computed rework quantity in primary quantity unit for counter values that were recorded during the period of the log record or rework quantity in primary quantity unit that was manually recorded during the period of the log record.

Open quantity (P)

Computed open quantity in primary quantity unit for counter values that were recorded during the period of the log record or open quantity in primary quantity unit that was manually recorded during the period of the log record.

Unit (P)

Primary quantity unit according to a configuration relating to workplaces/machines (resource configuration).

Yield (S), Scrap (S), Rework (S), Open quantity (S), Unit (S)

The corresponding quantities in secondary quantity unit are converted from the primary quantity according to the definitions made for conversions in the workplace configuration and provided that corresponding conversion factors are available.

Please consider the "convert quantities" option for manual quantity changes.

Yield (T), Scrap (T), Rework (T), Open quantity (T), Unit (T)

The corresponding quantities in tertiary quantity unit are converted from the primary quantity according to the definitions made for conversions in the workplace configuration and provided that corresponding conversion factors are available.

Please consider the "convert quantities" option for manual quantity changes.

Yield (B), Scrap (B), Rework (B), Open quantity (B), Unit (B)

The corresponding quantities in base quantity unit are converted from the primary quantity according to the definitions made for conversions in the workplace configuration and provided that corresponding conversion factors are available.

Please consider the "convert quantities" option for manual quantity changes.

Convert quantities

When quantities are changed, the "convert quantities" option allows for quantities to be converted according to the conversion factors defined at the machine or the configured formulas for quantity conversion.

If the conversion factors of active operations are to be used for the machine (compare the option "basis for HYDRA-MDE quantity conversion" in the workplace configuration), it is urgently recommended not to activate the "quantity conversion" option, when it comes to subsequent changes in the maintenance of postings function of HYDRA-MDE, as only conversion factors that are specific to machines are taken into account by the maintenance of postings function.

Cycles

The number of cycles that were recorded within the period of the log record.

Toolbar

Postings relating to orders

Function authorization: oboo.*
Starts the function "Order-related postings".

Transfers the workplace, cost center, date/time and shift from the selection area.